

8 Topics in Particle Cosmology

These notes cover a few selected topics relevant to particle cosmology, and are by absolutely no means complete. A familiarity with relativity and classical field theory, statistical mechanics, and quantum mechanics will be helpful, though the required results will be covered here. For deeper reading, I recommend the books:

- S. Weinberg, *Cosmology* (2008)
- D. Hooper, *Particle Cosmology & Astrophysics* (2024)
- E. W. Kolb and M. S. Turner, *The Early Universe* (1990)

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8.1 Introduction

Particle cosmology is a mix of astrophysics, cosmology, and particle physics. This is a very wide field, and we treat only a small selection of important examples. A rough timeline for particle cosmology is given in Table 8.1. We touch briefly on most of these events, and treat a few in more detail.

The concept at the heart of particle cosmology is that of particle thermodynamics. The *hot Big Bang* model is the idea that the universe began in a very hot, dense state. All Standard Model particles interact with each other to some degree (and, presumably, the same is true of particles beyond the Standard Model, such as dark matter). In quantum field theory, particles may be created and destroyed in such interactions. The high density and temperature of the early universe mean that these interactions, even if very weak, are enough to keep all particles in thermal equilibrium. As the universe expands, the particle density drops, and particles whose interactions are no longer fast enough fall out of thermal equilibrium. If these particles are stable (do not decay), their abundance is then “locked in” (a process termed *freeze out*). This links the particle interactions and the rate of expansion of the universe to observables, such as the temperature and density of the particles. This remarkably simple idea leads, for the most part, to excellent agreement with observations.

	t	T [GeV]	
Inflation*	10^{-36} – 10^{-32} s (?)	(?)	
Baryogenesis*	$\lesssim 10^{-11}$ s (?)	$\gtrsim 10^2$ (?)	
EW phase transition	$\sim 10^{-11}$ s	$\sim 10^2$	$T \sim m_{W^\pm}$
DM freeze-out*	10^{-11} – 10^{-5} s (?)	10^{-1} – 10^2 (?)	WIMPs
QCD phase transition	$\sim 10^{-5}$ s	$\sim 10^{-1}$	quarks $\rightarrow p, n, \pi$
Weak (ν) decoupling	~ 1 s	$\sim 10^{-3}$	$\Gamma_{\text{weak}} \lesssim H$
e^-/e^+ annihilation	~ 6 s	$\sim 5 \times 10^{-4}$	photon heating
BBN:			
p/n freeze-out	~ 1 s	$\sim 7 \times 10^{-4}$	$n/p \approx 1/6$
neutron decay	1–180 s		$n/p \rightarrow \sim 1/7$
He production	~ 3 min	$\sim 10^{-4}$	$Y_p \approx 25\%$
Matter domination	$\sim 5 \times 10^4$ yr	$\sim 8 \times 10^{-10}$	$\rho_m = \rho_r$
Recombination	$\sim 4 \times 10^5$ yr	$\sim 3 \times 10^{-10}$	$\Rightarrow$ CMB
Star/galaxy formation	$\gtrsim 10^8$ yr	$\lesssim 7 \times 10^{-12}$	
DE domination	$\sim 10^{10}$ yr	$\sim 3 \times 10^{-13}$	$\rho_\Lambda = \rho_m$
Today	1.4×10^{10} yr	2.3×10^{-13}	$T_0 = 2.7$ K

Table 8.1: A rough timeline of the universe. *Not fully understood.

8.1.1 Units

Since we are mixing relativity, particle physics and astrophysics, it is almost ubiquitous to use *natural units*, which are chosen so that the speed of light, c , the reduced Planck's constant $\hbar$, and the Boltzmann constant k_B , are set to one:

$$c = \hbar = k_B = 1. \quad (8.1)$$

Gravity brings one further constant, Newton's constant G , which we write in terms of the *Planck mass*, m_P , as

$$G = \frac{1}{m_P^2}. \quad (8.2)$$

Restoring $\hbar$ and c , the Planck mass is defined as¹

$$m_P \equiv \sqrt{\frac{\hbar c}{G}} \approx 1.22 \times 10^{19} \text{ GeV}/c^2. \quad (8.3)$$

We will use these natural units throughout, only restoring c or $\hbar$ factors if they help to make certain scalings more clear.

Each choice in Eq. (8.1) equates the dimensions of two quantities:

$$\begin{aligned} c = 1 : \quad & [\text{Length}] = [\text{Time}], \\ & [\text{Mass}] = [\text{Energy}], \\ \hbar = 1 : \quad & [\text{Energy}] = 1/[\text{Length}], \\ k_B = 1 : \quad & [\text{Energy}] = [\text{Temperature}]. \end{aligned}$$

Everything can therefore be measured in units of energy, for which we use GeV. Any quantity Q then has units

$$[Q] = \text{GeV}^n, \quad (8.4)$$

which defines its *mass dimension*, n . For example, masses, energies, and temperatures have $n = 1$, while lengths and times have $n = -1$. To convert to other units, we restore the appropriate combination of $\hbar$, c , k_B , and m_P , using

$$\hbar c \approx 197 \text{ MeV} \cdot \text{fm} \approx 2 \times 10^{-14} \text{ GeV} \cdot \text{cm}, \quad (8.5)$$

$$k_B \approx 8.6 \times 10^{-5} \text{ eV/K} \approx 10^{-10} \text{ MeV/K}, \quad (8.6)$$

$$c \approx 3 \times 10^{10} \text{ cm/s}. \quad (8.7)$$

¹A warning: it is also common to use the *reduced Planck mass*, $M_P = \frac{m_P}{\sqrt{8\pi}}$, particularly in relativistic cosmology since it simplifies the Friedmann equations. Unfortunately, not all sources use the same notation, and often the term ‘‘Planck mass’’, as well as the notations, m_P , M_P , $\bar{m}_P$ are all used interchangeably to mean either.

With $\hbar = c = 1$, the first of these relates energies to inverse lengths, while c relates lengths to times:

$$\text{GeV} \approx 5.07 \times 10^{13} \text{ cm}^{-1}, \quad (8.8)$$

Problem 8.1.1: Consider the product of a cross-section and a velocity, σv , which we shall need often. Show that σv has mass dimension -2 , and find the factor that converts it from GeV^{-2} to cm^3/s .

Solution (8.1.1): Dimensionally,

$$L^2 \times \frac{L}{T} \longleftrightarrow \frac{1}{\text{GeV}^2},$$

i.e., σv has mass dimension -2 . To go from GeV^{-2} to cm^3/s , we first make the quantity dimensionless by restoring the factors that were set to one (an $\hbar c$ for each length, and a c for the velocity), and then “multiply by 1” using Eq. (8.8) and the value of c :

$$\begin{aligned} \sigma v &\longrightarrow \frac{\sigma(v/c)}{(\hbar c)^2 \text{ GeV}^{-2}} \\ &= \frac{\sigma(v/c)}{(\hbar c)^2 \text{ GeV}^{-2}} \times \frac{c}{3 \times 10^{10} \text{ cm/s}} \times \frac{(5.07 \times 10^{13} \text{ cm}^{-1})^2 (\hbar c)^2}{\text{GeV}^2} \\ &= \frac{\sigma v}{\text{cm}^3/\text{s}} \times 8.57 \times 10^{16}, \end{aligned}$$

so that

$$\frac{\sigma v}{\text{cm}^3/\text{s}} = \left(\frac{\sigma v}{\text{GeV}^{-2}} \right) \times 1.17 \times 10^{-17}.$$

8.1.2 Review of special relativity

We shall need only a few results from special relativity, which we collect here, mostly to fix the notation; for details, see Classical Field Theory Secs. 2.1–2.3.

Space-time points, *events*, are labelled by four-vectors

$$x^\mu = (t, \mathbf{x}). \quad (8.9)$$

The main postulate is that the *interval* between two events,

$$ds^2 = dt^2 - d\mathbf{x}^2 = \eta_{\mu\nu} dx^\mu dx^\nu, \quad (8.10)$$

is *invariant* under Lorentz transformations, i.e., it is the same in all inertial frames². Here, $\eta_{\mu\nu}$ is the (flat space) *metric*,

$$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1). \quad (8.11)$$

The overall sign of the metric is a convention. There are two common choices. Our choice, which can be called the *mostly minus* metric, or the metric of signature -2 , is the standard choice for particle physics. The opposite choice, $\text{diag}(-1, 1, 1, 1)$ (signature $+2$), is standard in general relativity, and is used in

²We use the Einstein summation convention, in which repeated indices are summed over.

many cosmology texts. Since particle cosmology skirts both fields, extra care must be taken comparing formulas between sources. We use the particle physics convention throughout, as in Fields Sec. 2.2. The metric is used to lower indices, $a_\mu = \eta_{\mu\nu} a^\nu$, so that the product of two four-vectors,

$$a^\mu b_\mu = a^0 b^0 - \mathbf{a} \cdot \mathbf{b}, \quad (8.12)$$

is also invariant.

Inertial frames are related by Lorentz transformations. For a frame moving with velocity v along the x -axis (a *boost*),

$$t' = \gamma(t - vx), \quad (8.13)$$

$$x' = \gamma(x - vt), \quad (8.14)$$

with $y' = y$ and $z' = z$, where

$$\gamma \equiv \frac{1}{\sqrt{1 - v^2}}. \quad (8.15)$$

In matrix form, $x'^\mu = \Lambda^\mu_\nu x^\nu$, with

$$\Lambda = \begin{pmatrix} \gamma & -\gamma v & 0 & 0 \\ -\gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (8.16)$$

All four-vectors transform in the same way. We may also form more complicated objects such as *tensors*. A rank N tensor has N space-time indices, and transforms under Lorentz transformations with N copies of Λ :

$$T'^{\mu_1 \dots \mu_N} = \Lambda^{\mu_1}_{\nu_1} \dots \Lambda^{\mu_N}_{\nu_N} T^{\nu_1 \dots \nu_N}. \quad (8.17)$$

We write derivatives with respect to the space-time coordinates as

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu} = \left(\frac{\partial}{\partial t}, \nabla \right), \quad (8.18)$$

which transforms as a covariant vector (four-vector with a lower index). Raising the index with the metric changes the sign of the spatial components: $\partial^\mu \equiv \eta^{\mu\nu} \partial_\nu = \left(\frac{\partial}{\partial t}, -\nabla \right)$.

The four-velocity of a particle is

$$u^\mu \equiv \frac{dx^\mu}{d\tau} = (\gamma, \gamma \mathbf{v}), \quad (8.19)$$

where τ is the proper time ($d\tau = ds$), so that $u^\mu u_\mu = 1$. The energy and momentum of a particle of mass m form the *four-momentum* (see Fields Sec. 2.3),

$$p^\mu = m u^\mu = (E, \mathbf{p}), \quad (8.20)$$

whose invariant square is the mass:

$$p^\mu p_\mu = E^2 - \mathbf{p}^2 = m^2. \quad (8.21)$$

The velocity is $\mathbf{v} = \mathbf{p}/E$, and massless particles (e.g., photons) have $E = p$. Two limits will be used constantly:

$$E = \sqrt{p^2 + m^2} \approx \begin{cases} m + \frac{p^2}{2m} & \text{non-relativistic, } p \ll m, \\ p & \text{relativistic, } p \gg m. \end{cases} \quad (8.22)$$

For details and derivations, see Fields Secs. 2.1–2.3.

8.2 Principles of Cosmology

On the largest scales, the universe is observed to be homogeneous and isotropic (the *cosmological principle*), and electrically neutral, so that its dynamics are governed by gravity. A proper description requires general relativity, though most of the results we need follow from simple Newtonian arguments. This section is a brief review, included to keep these notes self-contained; for details see, e.g., Weinberg³.

In a homogeneous and isotropic universe, the only allowed overall motion is a uniform expansion (or contraction). The physical position of any point (e.g., a galaxy) can then be written as

$$x(t) = a(t) X, \quad (8.23)$$

where the *comoving* coordinate, X , is fixed, and the time dependence is put into the *scale factor*, $a(t)$. The distance between any two points is therefore

$$D(t) = \Delta x(t) = a(t) \Delta X, \quad (8.24)$$

and their relative velocity is

$$v = \frac{dD}{dt} = \dot{a}(t) \Delta X, \quad (8.25)$$

so that

$$\frac{v}{D} = \frac{\dot{a}}{a} \equiv H(t), \quad (8.26)$$

which defines the *Hubble parameter*. Evaluated today, with $a(t_0) \equiv 1$, this is *Hubble's law*:

$$v = H_0 D, \quad (8.27)$$

$$v \approx cz \quad (z \ll 1), \quad (8.28)$$

³S. Weinberg, *Cosmology* (2008).

with

$$H_0 \equiv 100 h \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (8.29)$$

where $H_0 = H(t_0)$ is the *Hubble constant*, and h is a dimensionless number, which keeps the dependence on the measured value of H_0 explicit. Observations put $h \approx 0.7$.

The wavelength of light is stretched by the expansion. The *redshift*, z , of light emitted at time t_{em} is defined as

$$1 + z \equiv \frac{\lambda_{\text{obs}}}{\lambda_{\text{em}}} = \frac{a(t_0)}{a(t_{\text{em}})} = \frac{1}{a(t_{\text{em}})}. \quad (8.30)$$

More generally, wavelengths, momenta, and the temperature of radiation scale with the expansion as

$$\lambda \propto a, \quad (8.31)$$

$$p \propto a^{-1}, \quad (8.32)$$

$$T \propto a^{-1}. \quad (8.33)$$

8.2.1 Newtonian cosmology

Consider a universe made of a roughly homogeneous collection of particles (e.g., galaxies). We consider the acceleration of one such particle at point P , as observed from some origin, O . According to the shell theorem, we draw a spherical shell that extends to the point P from the origin, taking it to be large enough that the mass density inside it may be treated as uniform. The contribution to the gravitational force from the uniform mass density inside the shell is equivalent to that of a point mass at the origin, with mass ρV , where ρ is the mass density, and V is the volume of the sphere. At the same time, the contribution from outside the shell cancels, and is zero. If, at a specific time, the galaxy is at a distance x , then

$$\ddot{x} = -\frac{GM_{\text{tot}}}{x^2} = -\frac{4\pi G}{3}\rho x. \quad (8.34)$$

Writing $x = a(t)X$,

$$\begin{aligned} \ddot{a}(t) X &= -\frac{4\pi G}{3}\rho(t) a(t) X \\ \implies \frac{\ddot{a}}{a} &= -\frac{4\pi G}{3}\rho(t). \end{aligned} \quad (8.35)$$

Notice, in particular, that no explicit position dependence remains in this formula! Since

$$\frac{d}{dt} \left(\frac{\dot{a}}{a} \right) = \frac{\ddot{a}}{a} - \left(\frac{\dot{a}}{a} \right)^2, \quad (8.36)$$

with $H \equiv \dot{a}/a$, we can write this as

$$\dot{H} = -H^2 - \frac{4\pi G}{3}\rho(t). \quad (8.37)$$

Now, if we keep the sphere locked onto the distance to the given galaxy, the size of the sphere is changing with time. However, the total mass inside the sphere is conserved,

$$M = \frac{4\pi}{3}\rho(t) a^3(t) X^3 = \text{const.} \quad (8.38)$$

$$\implies \rho \propto a^{-3}. \quad (8.39)$$

At the same time, the total energy (per unit mass) of the galaxy is also conserved,

$$\frac{1}{2}\dot{x}^2 - \frac{GM}{x} = U = \text{const.} \quad (8.40)$$

Substituting $x = aX$ and Eq. (8.38), this becomes

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2}, \quad (8.41)$$

where we defined

$$kc^2 \equiv -\frac{2U}{X^2}. \quad (8.42)$$

There are three cases, depending on the sign of the energy, which is set by the initial conditions of the system:

$$U = \begin{cases} -|U_0|, & k > 0 : \text{ bound (closed),} \\ 0, & k = 0 : \text{ critical (flat),} \\ +|U_0|, & k < 0 : \text{ unbound (open).} \end{cases} \quad (8.43)$$

In general relativity, k will describe the curvature of space. Homogeneity and isotropy allow only three geometries: positively curved (a three-sphere, $k > 0$), flat (Euclidean, $k = 0$), and negatively curved (hyperbolic, $k < 0$).

To continue, we will treat our homogeneous collection of particles as a gas or fluid, and make use of thermodynamics. For a fluid with pressure P , the first law of thermodynamics, $dE = -P dV$, applied to the expanding volume, gives the *fluid equation*:

$$\begin{aligned} d(\rho c^2 a^3) &= -P d(a^3) \\ \implies \dot{\rho} + 3H \left(\rho + \frac{P}{c^2} \right) &= 0. \end{aligned} \quad (8.44)$$

Here, and from now on, ρc^2 is the total energy density, not just that of the mass. To solve this, we need an *equation of state*, relating the pressure to the density,

$$P = w\rho c^2 \quad (8.45)$$

$$\implies \rho \propto a^{-3(1+w)}, \quad (8.46)$$

for constant w . The three important cases are

$$\rho = \begin{cases} \rho_0 a^{-3} & \text{matter, } w = 0, \\ \rho_0 a^{-4} & \text{radiation, } w = 1/3, \\ \rho_0 & \text{cosmological constant, } w = -1. \end{cases} \quad (8.47)$$

For *matter* (non-relativistic particles), the pressure is negligible, and the density is simply diluted with the volume. For *radiation* (photons, and any relativistic particles), the number density is also diluted as a^{-3} , but in addition the energy of each particle is redshifted, $E \propto a^{-1}$ [Eq. (8.31)], giving $\rho \propto a^{-4}$. Since $\rho \propto T^4$ for radiation, this is equivalent to $T \propto a^{-1}$. Radiation therefore dominates at early times, and matter at later times.

A component with $w = -1$ has constant energy density, and negative pressure, $P = -\rho c^2$. This is the equation of state of the *cosmological constant*, Λ (or *vacuum energy*), the simplest form of *dark energy*. Since it is not diluted by the expansion, it eventually dominates.

8.2.2 Friedmann equations

The Newtonian results carry over to general relativity (Sec. 8.2.3) with three changes: ρc^2 is the total energy density, pressure also gravitates, and a cosmological constant, Λ , is allowed. The *Friedmann equation* is

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}, \quad (8.48)$$

the *acceleration equation* is

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3P}{c^2}\right) + \frac{\Lambda c^2}{3}, \quad (8.49)$$

and the fluid equation is unchanged,

$$\dot{\rho} + 3H\left(\rho + \frac{P}{c^2}\right) = 0. \quad (8.50)$$

Only two of these three equations are independent.

The cosmological constant is equivalent to a fluid with constant density, and a negative pressure equal in magnitude to its energy density, i.e., $w = -1$ in Eq. (8.45):

$$\rho_\Lambda \equiv \frac{\Lambda c^2}{8\pi G}, \quad (8.51)$$

$$P_\Lambda = -\rho_\Lambda c^2. \quad (8.52)$$

From Eq. (8.49), the expansion accelerates ($\ddot{a} > 0$) when it is dominated by a component with $w < -1/3$.

The *critical density* is that for which the universe is flat ($k = 0$):

$$\rho_{\text{crit}} \equiv \frac{3H^2}{8\pi G}, \quad (8.53)$$

$$\begin{aligned} \rho_{\text{crit},0} &\approx 1.88 \times 10^{-26} \text{ h}^2 \text{ kg m}^{-3} \\ &\approx 1.05 \times 10^{-5} \text{ h}^2 \text{ GeV cm}^{-3}. \end{aligned} \quad (8.54)$$

Densities are usually quoted relative to this, as the *density parameters*, $\Omega_i \equiv \rho_i/\rho_{\text{crit}}$ (with $\Omega_k \equiv -kc^2/a^2H^2$ for the curvature). For a flat universe ($k = 0$) dominated by a single component, the solutions are

$$a(t) \propto \begin{cases} t^{2/3(1+w)} & w \neq -1, \\ t^{2/3} & \text{matter,} \\ t^{1/2} & \text{radiation,} \\ e^{Ht} & \Lambda. \end{cases} \quad (8.55)$$

Problem 8.2.1: Derive the solutions in Eq. (8.55) from the Friedmann equation, Eq. (8.48), for a flat universe containing a single component with $\rho \propto a^{-3(1+w)}$. Consider the case $w = -1$ separately, and show that H is then constant.

8.2.3 Link to general relativity

In general relativity, gravity is described by the metric of spacetime, $g_{\mu\nu}$, which replaces $\eta_{\mu\nu}$ in Eq. (8.10) (for a brief motivation, and for the geodesic equation below, see Fields Sec. 2.10). In this section, we set $c = 1$, and write $G = 1/m_P^2$. For the three geometries allowed by homogeneity and isotropy, the metric is, in spherical coordinates, with $d\Omega^2 \equiv d\theta^2 + \sin^2\theta d\phi^2$,

$$ds^2 = dt^2 - a^2(t) [dr^2 + r^2 d\Omega^2] \quad \text{flat,} \quad (8.56)$$

$$ds^2 = dt^2 - a^2(t) \left[\frac{dr^2}{1 - r^2/R^2} + r^2 d\Omega^2 \right] \quad \text{positively curved,} \quad (8.57)$$

$$ds^2 = dt^2 - a^2(t) \left[\frac{dr^2}{1 + r^2/R^2} + r^2 d\Omega^2 \right] \quad \text{negatively curved,} \quad (8.58)$$

where $a(t)R$ is the radius of curvature of space at time t . These are combined in the most general homogeneous and isotropic metric, the Friedmann-Lemaître-Robertson-Walker (FLRW) metric,⁴

$$ds^2 = dt^2 - a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right], \quad (8.59)$$

where k is the curvature of space ($k = 0, 1/R^2$, and $-1/R^2$ for the three cases above), corresponding to the three geometries of Eq. (8.43). It is also common to rescale the radial coordinate, $\tilde{r} \equiv r/R$, and the scale factor, $\tilde{a} \equiv aR$, so that

$$ds^2 = dt^2 - \tilde{a}^2(t) \left[\frac{d\tilde{r}^2}{1 - \kappa\tilde{r}^2} + \tilde{r}^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right], \quad (8.60)$$

where $\kappa = 0, +1$, or -1 for the flat, positively curved, and negatively curved cases, respectively. In this form, $\tilde{r}$ is dimensionless, and $\tilde{a}(t)$ is the radius of curvature, so that $\tilde{a}(t_0) \neq 1$.

Free particles (i.e., not under the influence of any force) move along *geodesics* of the metric,

$$\frac{du^\mu}{d\tau} + \Gamma_{\nu\lambda}^\mu u^\nu u^\lambda = 0, \quad (8.61)$$

where the Christoffel symbols, $\Gamma_{\nu\lambda}^\mu$, are built from derivatives of the metric (See Fields, or any GR text).

The metric is determined by the energy and momentum content of the universe, through the Einstein field equations,

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu} = \frac{8\pi}{m_P^2} T_{\mu\nu}, \quad (8.62)$$

where the Ricci tensor, $R_{\mu\nu}$, and Ricci scalar, $R = g^{\mu\nu} R_{\mu\nu}$, are built from the metric and its derivatives, and $T_{\mu\nu}$ is the stress-energy tensor (see Fields Sec. 2.5). Here, the cosmological constant is included in $T_{\mu\nu}$, as a fluid with constant density $\rho_\Lambda = \Lambda m_P^2/8\pi$ and pressure $P_\Lambda = -\rho_\Lambda$ [Eq. (8.51)]. It is often written explicitly instead, as an extra term, $+\Lambda g_{\mu\nu}$, on the right-hand side. Homogeneity and isotropy require that the stress-energy tensor has the form of a perfect fluid,

$$T_{\mu\nu} = (\rho + P) u_\mu u_\nu - P g_{\mu\nu}, \quad (8.63)$$

where u^μ is the four-velocity of the fluid (see Problem 8.2.2). In the local rest frame of the fluid, where $u^\mu = (1, 0, 0, 0)$ and $g_{\mu\nu} = \eta_{\mu\nu}$, this is

$$T_{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{pmatrix}. \quad (8.64)$$

⁴Recall that we use the $(+, -, -, -)$ sign convention for the metric (Sec. 8.1.2). Many cosmology texts, including Weinberg, use $(-, +, +, +)$, which changes the sign of some terms.

The components of Eq. (8.62) then give the equations of Sec. 8.2.2:

$$\begin{aligned} (00) &\longrightarrow \text{Friedmann equation, Eq. (8.48),} \\ (00) \ \&\ (ii) &\longrightarrow \text{acceleration equation, Eq. (8.49),} \\ \partial_\mu T^{\mu\nu} = 0 &\longrightarrow \text{fluid equation, Eq. (8.50).} \end{aligned}$$

For the derivations, see Weinberg.

Problem 8.2.2: In the rest frame of a perfect fluid, the stress-energy tensor has only an energy density, $T_{00} = \rho$, and an isotropic pressure, $T_{ij} = P \delta_{ij}$ (see Fields Sec. 2.5). Show that, in flat spacetime, the stress-energy tensor of a fluid moving with four-velocity u^μ is given by Eq. (8.63), with $g_{\mu\nu} = \eta_{\mu\nu}$.

Solution (8.2.2): The only rank-two tensors available are $u_\mu u_\nu$ and $\eta_{\mu\nu}$, so we must have

$$T_{\mu\nu} = A u_\mu u_\nu + B \eta_{\mu\nu},$$

where A and B are scalars. In the rest frame, $u_\mu = (1, 0, 0, 0)$, so that

$$T_{00} = A + B = \rho, \quad \text{and} \quad T_{ij} = -B \delta_{ij} = P \delta_{ij},$$

which gives $B = -P$ and $A = \rho + P$. Since this is a tensor equation, it holds in every frame. The same result follows by boosting the rest-frame tensor to the frame in which the fluid moves with velocity $\mathbf{v}$ (a boost by $-\mathbf{v}$), though this is more work. In curved spacetime, $\eta_{\mu\nu}$ is replaced by $g_{\mu\nu}$.

8.2.4 The Λ CDM model

The standard model of cosmology is the Λ CDM model: an FLRW universe containing radiation, ordinary (*baryonic*) matter, *cold dark matter* (CDM), and a cosmological constant, Λ . Here, “cold” means non-relativistic, so that the dark matter behaves as matter ($w = 0$). The present-day densities, are given in Table 8.2. What is typically measured most directly is the physical density, $\Omega_i h^2$. The total matter density is $\Omega_m = \Omega_c + \Omega_b$ (CDM and baryons).

		$\Omega_i h^2$	Ω_i
Baryons	Ω_b	0.0224 ± 0.0001	0.049
Cold dark matter	Ω_c	0.120 ± 0.001	0.264
Matter (total)	Ω_m	0.1428 ± 0.0011	0.315 ± 0.007
Radiation	Ω_r	4.2×10^{-5}	9.2×10^{-5}
Dark energy	Ω_Λ	0.311	0.6847 ± 0.0073
Curvature	Ω_k		0.001 ± 0.002

Table 8.2: Present-day density parameters of the Λ CDM model, from the Planck 2018 results⁶, with $h = 0.674 \pm 0.005$.

Since the components scale differently with a [Eq. (8.47)], the universe passes through three eras: *radiation domination* at early times, *matter domination* from $z \approx 3400$, and *dark energy domination* from $z \approx 0.3$. Most of what follows

takes place during radiation domination. For details, and for the observational evidence, see, e.g., Hooper⁷, or Kolb and Turner⁸.

8.3 Particle physics 101

8.3.1 The Standard Model

The *Standard Model* of particle physics is the theory of the known elementary particles and their interactions. It is a quantum field theory, in which each particle is an excitation of a corresponding field, and it describes the electromagnetic, weak, and strong interactions. It has been tested to high precision in laboratory experiments, though, as we shall see, it cannot account for all cosmological observations (Secs. 8.7 and 8.8).

The *fermions* (spin $\frac{1}{2}$) are the *leptons*, e^- , μ^- , τ^- , and the neutrinos, with masses up to $\sim \text{GeV}$, and the *quarks*, u , d , $\dots$, with masses $\sim \text{MeV} - 100 \text{ GeV}$; see Table 8.3. The *bosons* have spin 1 (or 0), and mediate the interactions between the fermions; see Table 8.5. The quarks are bound by the strong interaction into *hadrons*, such as the *baryons*, protons p and neutrons n , which have masses $\sim \text{GeV}$; see Table 8.4. Note that hadrons are not always spin $\frac{1}{2}$ (or even fermions!).

Each particle has an *antiparticle*, with the same mass and spin, and the opposite charge; it is written with a bar, e.g., $\bar{u}$ and $\bar{\nu}$, or with its charge, e.g., e^+ (the *positron*). Some neutral particles, such as the photon, are their own antiparticles. The spin and the antiparticles together determine the degeneracy, g (the number of internal states), of each species, which we shall need for the thermodynamics (Sec. 8.4.1). A massive particle of spin s has $2s + 1$ spin states, and a massless particle with spin has two; counting each particle together with its antiparticle, $g = 2$ for the photon, $g = 4$ for $e^\pm$, and $g = 2$ for each neutrino species (only one spin state of each neutrino and antineutrino is observed).

8.3.2 Interactions

Interactions between fermions (and hadrons, $\dots$) are mediated by the exchange of bosons, as in Fig. 8.1. This is an example of a *Feynman diagram*, which represents an amplitude, A , with

$$P \sim |A|^2, \quad (8.65)$$

where P is the probability for the process. Each vertex contributes a factor of the coupling strength, and the exchanged particle (the *propagator*) a factor that depends on its mass; in general, these factors are very complicated. Generally, one must consider *all* diagrams with the same initial and final states. For the

⁷D. Hooper, *Particle Cosmology & Astrophysics* (2024).

⁸E. W. Kolb and M. S. Turner, *The Early Universe* (1990).

			Generation	Mass	Charge	Spin
Quarks	Up	u	1	2 MeV	$+\frac{2}{3}$	$\frac{1}{2}$
	Down	d	1	5 MeV	$-\frac{1}{3}$	$\frac{1}{2}$
	Charm	c	2	1.3 GeV	$+\frac{2}{3}$	$\frac{1}{2}$
	Strange	s	2	93 MeV	$-\frac{1}{3}$	$\frac{1}{2}$
	Top	t	3	173 GeV	$+\frac{2}{3}$	$\frac{1}{2}$
	Bottom	b	3	4.2 GeV	$-\frac{1}{3}$	$\frac{1}{2}$
Leptons	Electron	e^-	1	0.511 MeV	-1	$\frac{1}{2}$
	Electron neutrino	ν_e	1	< 1 eV	0	$\frac{1}{2}$
	Muon	μ^-	2	106 MeV	-1	$\frac{1}{2}$
	Muon neutrino	ν_μ	2	< 1 eV	0	$\frac{1}{2}$
	Tau	τ^-	3	1.78 GeV	-1	$\frac{1}{2}$
	Tau neutrino	ν_τ	3	< 1 eV	0	$\frac{1}{2}$

Table 8.3: The fermions of the Standard Model. Each also has an antiparticle, with the opposite charge.

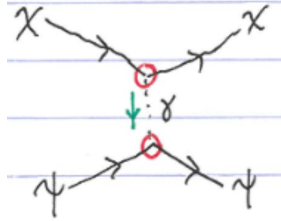


Figure 8.1: Feynman diagram: χ and ψ interact by exchanging a γ . The vertices and the propagator are marked.

most part, we will be able to treat this entire interaction phenomenologically, and introduce an “effective” vertex, as in Fig. 8.2.

Any number of in/out particles is allowed; “usually” (for us) it is $2 \rightarrow 2$ or $1 \rightarrow 2$. The effective vertex is either taken from theory, or treated as a parameter to be measured.

8.3.3 Scattering cross section, annihilation, and decay

We shall need three kinds of process: scattering, annihilation (and its reverse, production), and decay. In each case, the quantity of interest for cosmology is the rate at which the process occurs, since it is this that is compared with the expansion rate, H (Sec. 8.4). The rate is set by the particle density, together with a single quantity that contains the particle physics: the cross-section, σ , for

		Composition	Mass	Binding energy	Spin
Proton	p	uud	938 MeV		$\frac{1}{2}$
Neutron	n	udd	940 MeV		$\frac{1}{2}$
Pions	$\pi^\pm$	$u\bar{d}, d\bar{u}$	140 MeV		0
	π^0	$u\bar{u}, d\bar{d}$	135 MeV		0
Hydrogen	H	p	938 MeV		$\frac{1}{2}$
Deuterium	D (^2H)	pn	1876 MeV	2.2 MeV	1
Tritium	T (^3H)	$p2n$	2809 MeV	8.5 MeV	$\frac{1}{2}$
Helium	^4He	$2p2n$	3727 MeV	28.3 MeV	0

Table 8.4: Some common hadrons, and the lightest nuclei. The binding energy is the difference between the mass of the constituents and the mass of the bound state. It is given only for the nuclei: the masses of the hadrons are much larger than those of their quarks, so it is not meaningful there.

		Mass	Spin	Interaction
Photon	γ	0	1	EM
Weak bosons	$W^\pm$	80 GeV	1	weak
	Z	91 GeV	1	weak
Gluon	g	0	1	strong
Higgs	H	125 GeV	0	Higgs
Graviton (?)		0	2	gravity

Table 8.5: The bosons, which mediate the interactions. The graviton is hypothetical.

scattering and annihilation, and the decay rate, Γ , for decay.

Scattering. For *scattering*, the total four-momentum is conserved,

$$P_i^\mu = P_f^\mu, \quad (8.66)$$

so that

$$E_i = E_f, \quad (8.67)$$

$$\mathbf{p}_i = \mathbf{p}_f. \quad (8.68)$$

Angular momentum and the charges (electric charge, baryon number, lepton number) are also conserved. Scattering is *elastic* if the initial particles are the same as the final ones, and *inelastic* if there is a change in particle number or species.

The rate is set by the *scattering cross-section*. Take $2 \rightarrow N$,

$$A + B \rightarrow x, y, z \dots \quad (8.69)$$

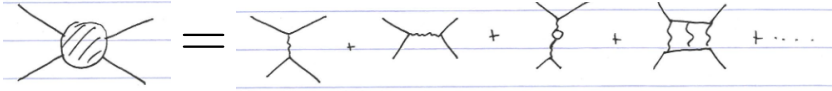


Figure 8.2: The sum of all diagrams (tree-level, loops, box, ...) is represented by a single “effective” vertex.

The rate per A particle is

$$R = n_B v_{\text{rel}} \times \sigma, \quad (8.70)$$

where n_B is the number density (number per volume), v_{rel} is the relative velocity, and σ is the cross-section, which contains all of the particle physics. The factor $n_B v_{\text{rel}}$ is the number of B 's passing through unit area each second, and $[R] = 1/[\text{Time}]$. The total rate is

$$R_T = n_B v \sigma N_A, \quad (8.71)$$

where N_A is the number of A particles, and the rate per unit volume is

$$R = n_A n_B \langle \sigma v \rangle. \quad (8.72)$$

Here, $\langle \sigma v \rangle$ is the *thermally-averaged cross-section*. In general, σ depends on the relative velocity, v , and the particles have a distribution of relative velocities, $f(v)$, so we average over it:

$$\langle \sigma v \rangle \equiv \int \sigma(v) v f(v) dv, \quad (8.73)$$

where $f(v)$ is normalised, $\int f(v) dv = 1$. More generally, the average is over the momentum distributions, $f(\mathbf{p})$, of the two species [see Eq. (8.88)]:

$$\langle \sigma v \rangle = \frac{\int \sigma v_{\text{rel}} f_A(\mathbf{p}_A) f_B(\mathbf{p}_B) d^3 p_A d^3 p_B}{\int f_A(\mathbf{p}_A) f_B(\mathbf{p}_B) d^3 p_A d^3 p_B}. \quad (8.74)$$

Annihilation (and production). Another important example is a special case of scattering called *annihilation*. A particle χ and its antiparticle $\bar{\chi}$ annihilate, as in Fig. 8.3. The cross-section and rates above apply to any scattering, including annihilation.

Relativistically, the reaction can go either way (annihilation or production), with

$$E_i = E_f, \quad (8.75)$$

where

$$E = \sqrt{p^2 + m^2}. \quad (8.76)$$

Non-relativistically, $E \approx m$, and the final states are constrained:

$$\sum_f m_f \leq 2m_\chi. \quad (8.77)$$

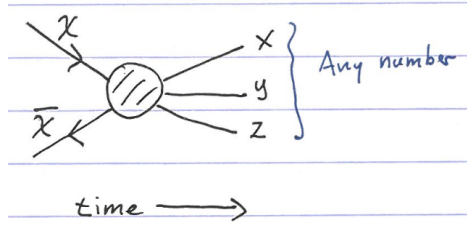


Figure 8.3: Annihilation, $\chi\bar{\chi} \rightarrow x, y, z$ (any number of final states). Time runs from left to right.

Decay. A particle may also *decay*, $1 \rightarrow N$, as in Fig. 8.4.

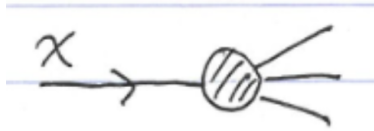


Figure 8.4: Decay of χ to several final-state particles.

This always requires

$$\sum_f m_f < m_\chi. \quad (8.78)$$

The rate per unit volume is

$$R = n_\chi \Gamma, \quad (8.79)$$

which defines the *decay rate*, Γ . The *lifetime* of the particle is

$$\tau \equiv \frac{1}{\Gamma}, \quad (8.80)$$

so that, in the absence of other processes, the number of particles falls as $e^{-t/\tau}$; the *half-life* is $t_{1/2} = \tau \ln 2$.

8.4 Cosmological thermodynamics

Scattering and decays allow energy exchange and particle creation. Elastic processes give *kinetic* equilibrium. Inelastic processes allow the species of particles to change, and allow *chemical* and kinetic equilibrium. Together, these give *thermal* equilibrium. Species can equilibrate if

$$R > H, \quad (8.81)$$

where

$$H = H(t), \quad (8.82)$$

$$R = R(n(t), T(t)) \quad (8.83)$$

are the expansion rate and the interaction rate per particle, both of which change as the universe expands and cools.

8.4.1 Equilibrium

For a species in thermal equilibrium at temperature T , the number density, energy density, and pressure⁹ follow from summing over the momentum states:

$$\begin{aligned} n &= \frac{g}{(2\pi)^3} \int f(p) \, d^3p \\ &= \frac{g}{2\pi^2} \int_0^\infty f(p) \, p^2 \, dp, \end{aligned} \quad (8.84)$$

$$\rho = \frac{g}{2\pi^2} \int_0^\infty f(p) \, E(p) \, p^2 \, dp, \quad (8.85)$$

$$P = \frac{g}{2\pi^2} \int_0^\infty f(p) \, \frac{p^2}{3E(p)} \, p^2 \, dp, \quad (8.86)$$

where

$$E(p) = \sqrt{p^2 + m^2}, \quad (8.87)$$

and $f(p)$ is the *distribution function* (the mean occupation of each state),

$$f(p) = \frac{1}{e^{(E-\mu)/T} \pm 1}. \quad (8.88)$$

Here g is the degeneracy (number of spin states), $+1$ is for fermions and -1 for bosons, and μ is the chemical potential, roughly the “cost” of changing the number of particles,

$$\mu \equiv \left(\frac{\partial U}{\partial N} \right)_{S,V}, \quad (8.89)$$

where U is the total energy.

There are two important limits (see Problems 8.4.1 and 8.4.2). In the *non-relativistic* limit, $T \ll m$, the ± 1 in Eq. (8.88) is negligible, so bosons and fermions

⁹The pressure is the momentum flux: a particle of momentum p moves with velocity p/E , and the factor of $\frac{1}{3}$ comes from averaging over directions.

behave in the same way, with

$$n = g \left(\frac{mT}{2\pi} \right)^{3/2} e^{-(m-\mu)/T}, \quad (8.90)$$

$$\rho = n \left(m + \frac{3}{2}T \right), \quad (8.91)$$

$$P = nT. \quad (8.92)$$

Each particle carries its rest energy and a thermal energy of $\frac{3}{2}T$, and the pressure is negligible, $P \ll \rho$: this is matter, $w = 0$ (Sec. 8.2.1). The $e^{-m/T}$ gives an exponential suppression: once T drops below m , the particles annihilate, and there is no longer enough thermal energy to produce them again.

In the *ultra-relativistic* limit, $T \gg m$, we have $E \approx p \sim T$, and (neglecting μ)

$$n = f_n \frac{\zeta(3)}{\pi^2} g T^3, \quad (8.93)$$

$$\rho = f_\rho \frac{\pi^2}{30} g T^4, \quad (8.94)$$

$$P = \frac{\rho}{3}, \quad (8.95)$$

with $\zeta(3) \approx 1.202$, and

$$f_n = \begin{cases} 1 & \text{bosons} \\ \frac{3}{4} & \text{fermions} \end{cases}, \quad f_\rho = \begin{cases} 1 & \text{bosons} \\ \frac{7}{8} & \text{fermions} \end{cases}. \quad (8.96)$$

This is radiation, $w = \frac{1}{3}$. Since there is no exponential suppression, the relativistic species dominate both the number and the energy density of the early universe.

In chemical equilibrium, for a reaction $A+B \leftrightarrow C+D$, the chemical potentials satisfy

$$\mu_A + \mu_B = \mu_C + \mu_D, \quad (8.97)$$

which can be used to eliminate the μ 's. The number of photons is not conserved (e.g., $e^- + \gamma \leftrightarrow e^- + 2\gamma$), so

$$\mu_\gamma = 0. \quad (8.98)$$

It then follows from $X + \bar{X} \leftrightarrow 2\gamma$ that a particle and its antiparticle have opposite chemical potentials, $\mu_{\bar{X}} = -\mu_X$: a non-zero μ corresponds to an excess of particles over antiparticles. In the early universe this excess is tiny (Sec. 8.8.1), so that $\mu \ll T$ for all relativistic species; in particular, we take

$$\mu_\nu \approx 0. \quad (8.99)$$

Finally, we define the *entropy density*, $s \equiv S/V$. For $\mu \ll T$, the first law, $T dS = d(\rho V) + P dV$, gives

$$s = \frac{\rho + P}{T}. \quad (8.100)$$

For a relativistic species, $P = \rho/3$, so that

$$s = \frac{4\rho}{3T} = f_\rho \frac{2\pi^2}{45} g T^3, \quad (8.101)$$

while the entropy of the non-relativistic species is negligible. The expansion is adiabatic, so the entropy in a comoving volume is conserved:

$$s a^3 = \text{const.} \quad (8.102)$$

Problem 8.4.1: Derive the non-relativistic number and energy densities, Eqs. (8.90) and (8.91), from Eqs. (8.84) and (8.85).

Solution (8.4.1): For $T \ll m$, the ± 1 in Eq. (8.88) is negligible, and $E \approx m + p^2/2m$, so that

$$f(p) \approx e^{-(m-\mu)/T} e^{-p^2/2mT}.$$

With $p = \sqrt{2mT} x$, the integrals are Gaussian:

$$n = \frac{g}{2\pi^2} e^{-(m-\mu)/T} (2mT)^{3/2} \int_0^\infty x^2 e^{-x^2} dx = g \left(\frac{mT}{2\pi} \right)^{3/2} e^{-(m-\mu)/T},$$

since the integral is $\sqrt{\pi}/4$. For ρ , the integrand has the extra factor $E \approx m + T x^2$; since $\int_0^\infty x^4 e^{-x^2} dx = 3\sqrt{\pi}/8$, the second term gives $\frac{3}{2}T$ per particle, and $\rho = n(m + \frac{3}{2}T)$.

Problem 8.4.2: Derive the ultra-relativistic number and energy densities, Eqs. (8.93) and (8.94), for both bosons and fermions. You may use

$$\int_0^\infty \frac{x^2}{e^x - 1} dx = 2\zeta(3), \quad \int_0^\infty \frac{x^3}{e^x - 1} dx = \frac{\pi^4}{15}.$$

Solution (8.4.2): With $E \approx p$, $\mu \approx 0$, and $p = Tx$,

$$n = \frac{g}{2\pi^2} T^3 \int_0^\infty \frac{x^2}{e^x \pm 1} dx, \quad \rho = \frac{g}{2\pi^2} T^4 \int_0^\infty \frac{x^3}{e^x \pm 1} dx.$$

For bosons, the given integrals lead directly to Eqs. (8.93) and (8.94), with $f_n = f_\rho = 1$. For fermions, we use

$$\frac{1}{e^x + 1} = \frac{1}{e^x - 1} - \frac{2}{e^{2x} - 1},$$

and rescale $x \rightarrow x/2$ in the second term, so that the integral of $x^k/(e^x + 1)$ is $1 - 2^{-k}$ times the boson one: $f_n = \frac{3}{4}$ ($k=2$) and $f_\rho = \frac{7}{8}$ ($k=3$).

8.4.2 Radiation domination

The energy densities of matter and radiation scale differently with the expansion (Sec. 8.2.1),

$$\rho_m \sim a^{-3}, \quad (8.103)$$

$$\rho_r \sim a^{-4}, \quad (8.104)$$

so that, going back in time, the radiation becomes more and more important. For $T \gtrsim \text{eV}$, it dominates the energy density, and therefore the expansion: this

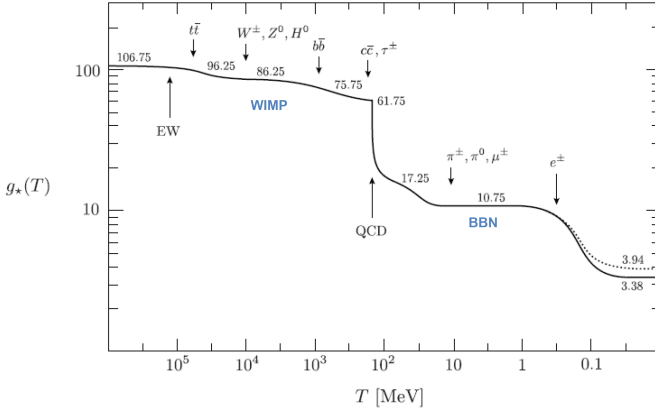


Figure 8.5: The effective relativistic degrees of freedom, $g_*(T)$, from $T \sim \text{TeV}$ down to eV. The dotted line shows the effective *entropic* degrees of freedom, h_{*S} . From Ref. [5].

is the *radiation-dominated* era. For a flat universe (Sec. 8.2.2), the Friedmann equation is

$$\left(\frac{\dot{a}}{a}\right)^2 = H^2 = \frac{8\pi G}{3}\rho = \frac{8\pi}{3m_P^2}\rho, \quad (8.105)$$

which is a good approximation at these early times, when the curvature and Λ terms are negligible.

Here, “radiation” means all of the relativistic species. Summing Eq. (8.94) over these, the total energy density is

$$\rho = g_* \frac{\pi^2}{30} T^4, \quad (8.106)$$

where T is the temperature of the photons, and g_* is the *effective number of relativistic degrees of freedom*,

$$g_*(T) \equiv g_{\text{eff}}^{\text{rel}} = \sum_b g_b \left(\frac{T_b}{T}\right)^4 + \frac{7}{8} \sum_f g_f \left(\frac{T_f}{T}\right)^4. \quad (8.107)$$

The sums are over the relativistic bosons (b) and fermions (f), and the temperature ratios appear because not all species need be in thermal equilibrium with the photons. For example, at $T \sim \text{MeV}$ the relativistic species are the photons ($g = 2$), $e^\pm$ ($g = 4$), and the three neutrinos ($g = 6$), all at the same temperature, which gives $g_* = 2 + \frac{7}{8} \times 10 = 10.75$.

As T drops, particles become non-relativistic when $T \sim m$; annihilation continues, $n \sim e^{-m/T}$, and the species no longer contributes to g_* . Therefore, g_* decreases in steps as the universe cools, as shown in Fig. 8.5.

With Eq. (8.106), the Friedmann equation gives the expansion rate as a function of the temperature,

$$H(T) = \sqrt{\frac{8\pi^3 g_*(T)}{90}} \frac{T^2}{m_P} \approx 1.66\sqrt{g_*} \frac{T^2}{m_P}. \quad (8.108)$$

For later use, we note that the critical density today, Eq. (8.53), is

$$\rho_{\text{crit}} = \frac{3H_0^2}{8\pi G} \approx 10^{-5} h^2 \frac{\text{GeV}}{\text{cm}^3} \approx 8 \times 10^{-47} h^2 \text{GeV}^4. \quad (8.109)$$

During radiation domination, $a \propto t^{1/2}$ [Eq. (8.55)], so that $H = 1/2t$, and the age of the universe at temperature T is

$$t \approx 2.4 \text{ s} \frac{1}{\sqrt{g_*}} \left(\frac{\text{MeV}}{T} \right)^2. \quad (8.110)$$

Similarly, summing Eq. (8.101) over the relativistic species, the total entropy density is

$$s = \sum_i \frac{\rho_i + P_i}{T_i} = \frac{2\pi^2}{45} h_*(T) T^3, \quad (8.111)$$

where h_* counts the entropic degrees of freedom,

$$h_*(T) = \sum_b g_b \left(\frac{T_b}{T} \right)^3 + \frac{7}{8} \sum_f g_f \left(\frac{T_f}{T} \right)^3, \quad (8.112)$$

with $h_* \approx g_*$ (they are equal when all species have the same temperature). Since the entropy in a comoving volume is conserved, Eq. (8.102), we have $h_* T^3 a^3 = \text{const}$: the temperature falls as $T \propto a^{-1}$, except when h_* changes.

8.4.3 Freeze-out and decoupling

In equilibrium, non-relativistic species are exponentially suppressed [Eq. (8.90)],

$$n \sim e^{-m/T}, \quad (8.113)$$

$$\rho \sim mn, \quad (8.114)$$

while relativistic species are not [Eqs. (8.93) and (8.94)],

$$n \sim T^3, \quad (8.115)$$

$$\rho \sim T^4. \quad (8.116)$$

The temperature today is

$$T_{\text{today}} \sim 3 \text{ K} \sim 10^{-4} \text{ eV}, \quad (8.117)$$

so *all* species are non-relativistic, except γ .¹⁰ If every species had remained in equilibrium, there would be essentially nothing left but the photons. So, why is the universe not just photons?

Equilibrium is maintained by the interactions, which must be fast compared to the expansion, Eq. (8.81). A species cannot remain in equilibrium if

$$R < H; \quad (8.118)$$

it then “decouples” from the thermal bath. Here, R is the interaction rate per particle [Eq. (8.70)],

$$R = n\langle\sigma v\rangle, \quad (8.119)$$

where n is the number density of the particles it interacts with. The condition $R < H$ means that a particle interacts, on average, less than once in the age of the universe, $t \sim 1/H$.

As an example, consider two species, χ and ψ , with $m_\chi \gg m_\psi$, which are kept in equilibrium by

$$\chi + \bar{\chi} \rightleftharpoons \psi + \bar{\psi}, \quad (8.120)$$

where, for example, $\psi = \gamma$. For $T \gg m_\chi$, the reaction proceeds freely in both directions. Once $T < m_\chi$, few of the ψ have enough energy to produce a $\chi\bar{\chi}$ pair, while the annihilation continues, so that n_χ follows the falling equilibrium value, $n_{\text{eq}} \sim e^{-m_\chi/T}$. However, as n_χ falls, so does the annihilation rate, $R = n_\chi\langle\sigma v\rangle$, until $R \sim H$, as shown in Fig. 8.6.

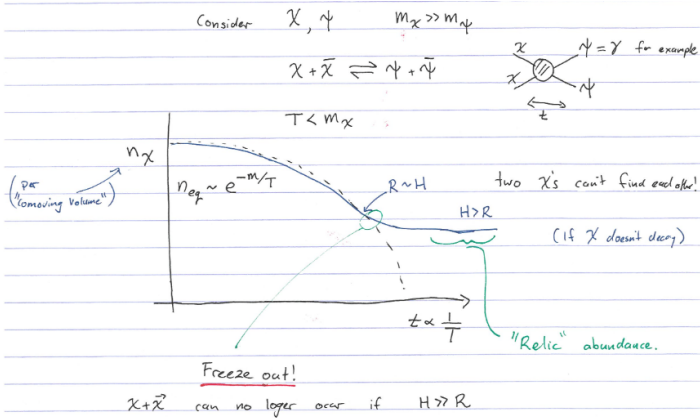


Figure 8.6: Freeze-out: n_χ (per comoving volume) against t , for $T < m_\chi$. It follows $n_{\text{eq}} \sim e^{-m/T}$ until $R \sim H$, then flattens to the “relic” abundance for $H > R$.

¹⁰The lightest neutrino may also still be relativistic.

Freeze-out. $\chi + \bar{\chi}$ annihilation can no longer occur once $H \gg R$: the two χ 's cannot find each other! The number of χ per comoving volume is then fixed, and we say that the abundance has *frozen out*. If χ does not decay, this leaves a “relic” abundance, which survives until today. Note that

$$R \propto \sigma n, \quad (8.121)$$

so a larger σ keeps the species in equilibrium for longer. This implies a later freeze-out, further down the falling equilibrium curve, and therefore *fewer* particles remaining. This is crucial for dark matter.

8.4.4 The Boltzmann equation

To follow a species as it leaves equilibrium, we need an equation for the evolution of its number density. Consider first *no* interactions, and let N be the number of particles per comoving volume,

$$N = a^3 n = \text{const.} \quad (8.122)$$

Then $dN/dt = 0$ gives

$$\frac{d(a^3 n)}{dt} = 3a^2 \dot{a} n + a^3 \dot{n} = 0 \quad (8.123)$$

$$\implies 3\frac{\dot{a}}{a}n + \dot{n} = 0, \quad (8.124)$$

so

$$\frac{dn}{dt} + 3Hn = 0. \quad (8.125)$$

This is just the dilution of the particles by the expansion, $n \propto a^{-3}$. Adding interactions gives the *Boltzmann equation*,¹¹

$$\frac{dn_i}{dt} + 3Hn_i = C_i(n_j), \quad (8.126)$$

where C_i is the “collision” term: the net rate, per unit volume, at which particles of species i are created. It depends on the densities, n_j , of the species that i interacts with, and has the form

$$C = R_{\text{production}} - R_{\text{annihilation}} - R_{\text{decay}}, \quad (8.127)$$

where the R 's are now rates per unit volume [Eq. (8.72)].

The behaviour depends on the relative size of the two sides of Eq. (8.126). If the interactions are fast compared to the expansion, the collision term dominates, and drives the density to its equilibrium value, for which $C = 0$ (production balances destruction). If they are slow, the collision term is negligible, and the particles are simply diluted, as in Eq. (8.125), and the species has frozen out.

¹¹Strictly, the Boltzmann equation is for the distribution function, $f(p, t)$; Eq. (8.126) is its integral over the momenta.

8.4.5 Thermal dark matter production: the WIMP miracle

We now apply the Boltzmann equation to dark matter (Sec. 8.7). Suppose that the dark matter is a new massive particle, χ , which can annihilate into (and be produced from) lighter Standard Model particles, ψ ,

$$\chi + \bar{\chi} \longleftrightarrow \psi + \bar{\psi}, \quad (8.128)$$

with $m_\chi > m_\psi$, as in Fig. 8.7. Such a particle is called a *WIMP* (weakly interacting massive particle). There is no decay, since dark matter must be stable.

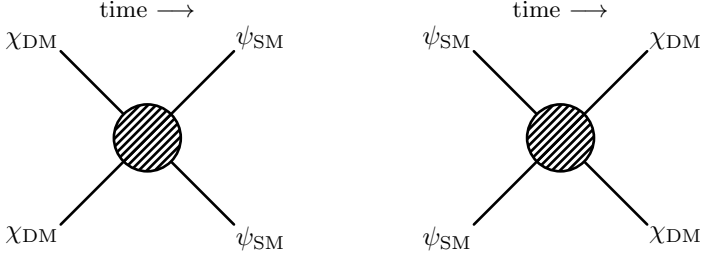


Figure 8.7: Annihilation (left), and production (right) $\chi\bar{\chi} \leftrightarrow \psi_{\text{SM}}\bar{\psi}_{\text{SM}}$ of a heavy dark matter particle χ into standard model particle ψ , via an effective vertex.

The annihilation rate per unit volume is

$$R_{\text{ann}} = \langle \sigma v \rangle n_\chi^2, \quad (8.129)$$

since $n_{\bar{\chi}} = n_\chi$. The production rate does not depend on n_χ : it is set by the ψ , which remain in equilibrium with the thermal bath. In equilibrium, production must balance annihilation ($dN/dt = 0$), which fixes

$$R_{\text{prod}} = \langle \sigma v \rangle n_{\text{eq}}^2, \quad (8.130)$$

where n_{eq} is the equilibrium density of χ . The Boltzmann equation, with $n = n_\chi$, is therefore

$$\frac{dn}{dt} = -3Hn + \langle \sigma v \rangle (n_{\text{eq}}^2 - n^2). \quad (8.131)$$

Practical calculation details. In practice, it is easier to work in dimensionless and comoving variables (the details are not important). We define

$$s = \frac{S}{a^3}, \quad (8.132)$$

$$Y = \frac{n}{s}, \quad (8.133)$$

$$x = \frac{m}{T} \propto t^{1/2}, \quad (8.134)$$

where s is the entropy density (S is the entropy per comoving volume, which is conserved; Sec. 8.4.1), and Y is the comoving abundance. That is, Y is proportional to the number of χ per comoving volume, and x plays the role of time. With

$$s = \frac{\rho + P}{T} = \frac{2\pi^2}{45} g_* T^3, \quad (8.135)$$

$$H \approx 1.66 \sqrt{g_*} \frac{T^2}{m_P}, \quad (8.136)$$

where $g_* \approx 86.25$ is the effective number of relativistic degrees of freedom at the temperatures of interest (Fig. 8.5), Eq. (8.131) becomes

$$\frac{dY}{dx} = \left(\sqrt{\frac{\pi}{45}} m_P \sqrt{g_*} \right) \frac{m}{x^2} \langle \sigma v \rangle (Y_{\text{eq}}^2(x) - Y^2(x)). \quad (8.137)$$

The dilution term has been absorbed into Y , which changes only through the interactions. In the simple case (Maxwell-Boltzmann statistics),

$$Y_{\text{eq}}(x) \approx \frac{45}{4\pi^4} \frac{x^2}{g_*} (2J_\chi + 1) K_2(x), \quad (8.138)$$

where K_2 is the modified Bessel function of the second kind, and $J_\chi = \frac{1}{2}$ is the spin. At late times Y is constant, and its value today gives the dark matter density, $\rho_\chi = m_\chi Y_{\text{now}} s_{\text{now}}$. To convert $Y \rightarrow \Omega h^2$,

$$Y_{\text{now}} \approx 3.63 \times 10^{-9} \left[\frac{\text{GeV}}{m_\chi} \right] \Omega h^2. \quad (8.139)$$

The miracle. The solutions of Eq. (8.137) are shown in Fig. 8.8. At early times, Y follows Y_{eq} ; it then freezes out at $x_f \approx 25$, almost independently of the parameters, and is constant thereafter. The larger the cross-section, the later the freeze-out, and the smaller the relic abundance.

We can understand this from a simple estimate. Freeze-out occurs when $n\langle\sigma v\rangle \approx H$, after which Y is constant, so that

$$Y_{\text{now}} \approx \frac{H(T_f)}{\langle\sigma v\rangle s(T_f)} \approx \frac{3.8 x_f}{\sqrt{g_*} m_P m_\chi \langle\sigma v\rangle}, \quad (8.140)$$

where $T_f = m_\chi/x_f$ is the freeze-out temperature. With Eq. (8.139), the mass drops out, and

$$\Omega h^2 \sim 0.1 \times \frac{3 \times 10^{-26} \text{ cm}^3/\text{s}}{\langle\sigma v\rangle}. \quad (8.141)$$

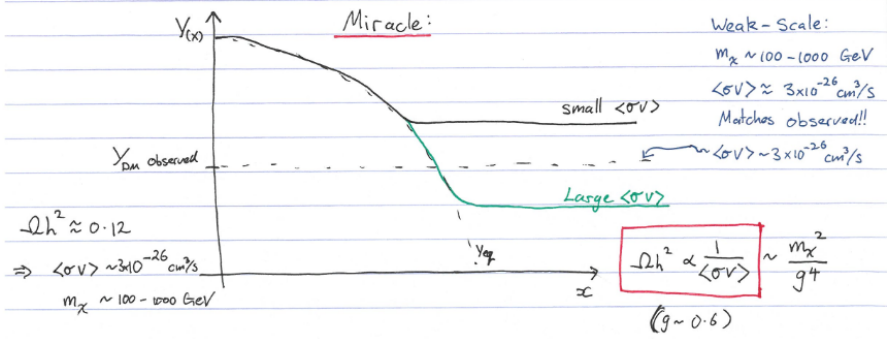


Figure 8.8: $Y(x)$ against $x: Y_{\text{eq}}$, and the solutions for small $\langle\sigma v\rangle$ (freezing out above the observed Y_{DM}) and large $\langle\sigma v\rangle$ (freezing out below it). The observed abundance corresponds to $\langle\sigma v\rangle \sim 3 \times 10^{-26} \text{ cm}^3/\text{s}$.

That is, the observed abundance, $\Omega h^2 \approx 0.12$, implies $\langle\sigma v\rangle \sim 3 \times 10^{-26} \text{ cm}^3/\text{s}$.

Now, for a particle of mass m_χ that interacts with coupling g , we expect $\langle\sigma v\rangle \sim \alpha^2/m_\chi^2$, with $\alpha = g^2/4\pi$, so that

$$\Omega h^2 \propto \frac{1}{\langle\sigma v\rangle} \sim \frac{m_\chi^2}{g^4}. \quad (8.142)$$

At the weak scale,

$$m_\chi \sim 100\text{--}1000 \text{ GeV}, \quad (8.143)$$

$$g \sim 0.6, \quad (8.144)$$

this gives $\langle\sigma v\rangle \sim 10^{-26}\text{--}10^{-24} \text{ cm}^3/\text{s}$, which matches the observed abundance! This coincidence is the *WIMP miracle*: a new stable particle at the weak scale would be produced with just the abundance needed to be the dark matter.

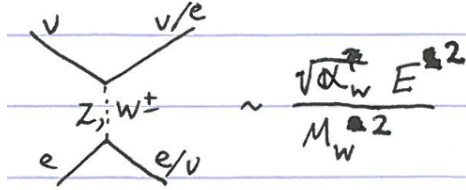
8.5 Weak decoupling and recombination

8.5.1 Weak (and neutrino) decoupling

The weak interaction, and with it the neutrinos, decouples at $T \sim \text{MeV}$. This is well after the EW phase transition,

$$T \ll M_W, \quad (8.145)$$

so the $W^\pm$ and Z are too heavy to be produced, and appear only as the exchanged (virtual) particles, as in Fig. 8.9.

Figure 8.9: ν - e scattering via $Z, W^\pm$ exchange.

Then

$$\sigma\nu \sim (\text{diagram})^2 \sim G_F^2 T^2, \quad (8.146)$$

where G_F is the *Fermi constant*,

$$G_F \sim \frac{\alpha_w}{M_W^2} \approx 1.17 \times 10^{-5} \text{ GeV}^{-2}. \quad (8.147)$$

Here, α_w is the weak coupling, and the factor of $1/M_W^2$ comes from the propagator of the heavy W , which suppresses the interaction for $E \ll M_W$. The factor of T^2 then follows on dimensional grounds, since the only other scale is the energy, $E \sim T$.

Neutrinos are relativistic, and only interact through the weak interaction:

$$\nu_e + \bar{\nu}_e \longleftrightarrow e^- + e^+, \quad (8.148)$$

$$e^- + \bar{\nu}_e \longleftrightarrow e^- + \bar{\nu}_e, \quad (8.149)$$

$$e^- + \nu \longleftrightarrow e^- + \nu. \quad (8.150)$$

These reactions keep the neutrinos in equilibrium with the $e^\pm$, and so with the photons. Then, from Eqs. (8.93) and (8.94),

$$\rho_\nu = \frac{7}{8} \frac{\pi^2}{30} g T^4, \quad (8.151)$$

with $g = 3 \times 2$ for $\nu, \bar{\nu}$ of each of (e, μ, τ) , and

$$n_\nu = \frac{3}{4} \frac{\zeta(3)}{\pi^2} 6 T^3. \quad (8.152)$$

With $g_* \approx 10.75$ for (e^-, e^+, γ, ν) , the expansion rate, Eq. (8.108), is

$$H = 1.66 \sqrt{g_*} \frac{T^2}{m_P} \sim 5 \frac{T^2}{m_P}. \quad (8.153)$$

The condition to leave equilibrium is (with $v \sim 1$)

$$R \approx n \langle \sigma v \rangle \sim H, \quad (8.154)$$

where, since $n \sim T^3$ for relativistic species,

$$n \langle \sigma v \rangle \sim G_F^2 T^5. \quad (8.155)$$

The weak rate therefore falls much faster than the expansion rate as the universe cools. The two are equal when $T^3 \sim 5/G_F^2 m_P$, which gives

$$T \sim \text{MeV}. \quad (8.156)$$

Therefore, neutrinos and photons are no longer in thermal equilibrium after $T_\gamma \sim \text{MeV}$.

After this, no other relativistic species interacts with ν (the non-relativistic p/n are too few to impact the thermodynamics of ν), so the neutrinos are simply diluted,

$$n_\nu \sim a^{-3}. \quad (8.157)$$

Their momenta are redshifted, $p \propto a^{-1}$, so they keep a thermal distribution, with $T_\nu \propto a^{-1}$. This is the same scaling as for the photons. If nothing else changed T_γ , then, even though decoupled, we would have

$$T_\gamma \approx T_\nu. \quad (8.158)$$

However, the photons are subsequently heated by the e^-/e^+ annihilation.

8.5.2 e^-/e^+ annihilation and photon reheating

For $T \lesssim 0.5 \text{ MeV}$, with $m_e = 0.5 \text{ MeV}$, the electrons and positrons become non-relativistic, and annihilate,

$$e^- + e^+ \rightarrow 2\gamma, \quad (8.159)$$

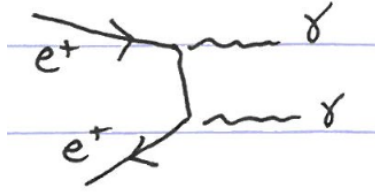
so energy is injected into the photons. The neutrinos have already decoupled, and are not heated: the photons end up hotter than the neutrinos.

The temperatures follow from entropy conservation, $h_* T^3 a^3 = \text{const}$ (Sec. 8.4.2), applied to the species in equilibrium with the photons. Before the annihilation, these are γ and $e^\pm$, with $h_* = 2 + \frac{7}{8} \times 4 = \frac{11}{2}$; afterwards, only γ , with $h_* = 2$. Therefore,

$$\frac{11}{2} (aT)_{\text{before}}^3 = 2 (aT_\gamma)_{\text{after}}^3, \quad (8.160)$$

while aT_ν is constant for the neutrinos, with $T_\nu = T$ before the annihilation. This gives

$$T_\nu \approx \left(\frac{4}{11} \right)^{1/3} T_\gamma, \quad (8.161)$$

Figure 8.10: Annihilation, $e^-e^+ \rightarrow \gamma\gamma$.

and, from Eq. (8.107),

$$g_* = 2 + \frac{7}{8} \times 2 N_{\text{eff}} \left(\frac{4}{11} \right)^{4/3} \approx 3.36. \quad (8.162)$$

Here, $N_{\text{eff}} = 3$ if the neutrinos decoupled instantly. But some e^-/e^+ energy went into ν , so that N_{eff} is slightly larger, as we discuss next.

8.5.3 The effective number of neutrino species

After the e^-/e^+ annihilation, the radiation consists of the photons and the colder, decoupled neutrinos. It is conventional to write its energy density as

$$\rho_r = \rho_\gamma \left[1 + \frac{7}{8} \left(\frac{4}{11} \right)^{4/3} N_{\text{eff}} \right], \quad (8.163)$$

which defines the *effective number of neutrino species*, N_{eff} , and is equivalent to Eq. (8.162). From Eqs. (8.94) and (8.161), each neutrino species (ν and $\bar{\nu}$, so $g = 2$) that decoupled instantly contributes exactly one unit, giving $N_{\text{eff}} = 3$. However, the neutrinos were not completely decoupled when the e^-/e^+ annihilated, and so were slightly heated. The full Standard Model calculation¹² gives

$$N_{\text{eff}} = 3.044. \quad (8.164)$$

Note that N_{eff} counts more than neutrinos: *any* light species that is relativistic at this time adds to ρ_r , whether or not it interacts with ordinary matter. Such “dark radiation” (e.g., sterile neutrinos or light axions) would increase N_{eff} above this value. For a review of such light relics, and their effect on the CMB, see Baumann¹³. It can be measured because ρ_r sets the expansion rate, Eq. (8.105):

¹²J. J. Bennett, G. Buldgen, P. F. de Salas, et al., *Towards a precision calculation of N_{eff} in the Standard Model II: Neutrino decoupling in the presence of flavour oscillations and finite-temperature QED*, JCAP **04**, 073 (2021).

¹³D. Baumann, *TASI lectures on primordial cosmology*, arXiv:1807.03098.

a larger N_{eff} means a faster expansion, and so an earlier p/n freeze-out and more helium (Sec. 8.6.1); it also leaves an imprint on the CMB. The Planck measurement¹⁴,

$$N_{\text{eff}} = 2.99 \pm 0.17, \quad (8.165)$$

agrees with the Standard Model value, and places tight constraints on additional light particles.

8.5.4 Hot relics and the neutrino background

The neutrinos decoupled while they were still relativistic: they are a *hot relic*. Unlike for a cold relic (Sec. 8.4.5), their abundance is not suppressed by $e^{-m/T}$, and their number density simply follows that of the photons. From Eqs. (8.93) and (8.161), with $T_\gamma = 2.7$ K today, there is a *cosmic neutrino background* with

$$T_\nu = \left(\frac{4}{11}\right)^{1/3} T_\gamma \approx 1.95 \text{ K}, \quad (8.166)$$

$$n_\nu = \frac{3}{11} n_\gamma \approx 112 \text{ cm}^{-3}, \quad (8.167)$$

for each species (ν and $\bar{\nu}$).

Since neutrinos have mass, they are (mostly) non-relativistic today, with $\rho_\nu = n_\nu \sum m_\nu$, so that

$$\Omega_\nu h^2 \approx \frac{\sum m_\nu}{94 \text{ eV}}. \quad (8.168)$$

The neutrinos are therefore a form of dark matter, but they cannot be all of it. That would require $\sum m_\nu \approx 11 \text{ eV}$; more importantly, hot dark matter moves too fast to clump on small scales, and would erase the structure that we observe. Conversely, the observed structure gives the strongest limit on the neutrino masses¹⁵,

$$\sum m_\nu < 0.12 \text{ eV}. \quad (8.169)$$

8.5.5 Recombination

Much later, at $T \sim \text{eV}$, the electrons and protons combine to form neutral hydrogen. The reaction is



with

$$\Delta E \approx 13 \text{ eV}, \quad (8.171)$$

¹⁴Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, Astron. Astrophys. **641**, A6 (2020).

¹⁵Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, Astron. Astrophys. **641**, A6 (2020).

where $\Delta E = m_e + m_p - m_H$ is the binding energy of hydrogen. Since $T \sim \text{eV} \ll m_e, m_p, m_H$, all three are non-relativistic, and Eq. (8.90) gives

$$n_i^{\text{eq}} = g_i \left(\frac{m_i T}{2\pi} \right)^{3/2} \exp([\mu_i - m_i]/T). \quad (8.172)$$

With $\mu_\gamma = 0$, we eliminate the μ 's using [Eq. (8.97)]

$$\mu_e + \mu_p = \mu_H, \quad (8.173)$$

and take $n_e = n_p$, since the universe is neutral. The relevant masses and degeneracies are:

	m	g
e^-	0.5 MeV	2
p	GeV	2
H	GeV	4

Then

$$\left(\frac{n_H}{n_e n_p} \right)_{\text{eq}} = \frac{g_H}{g_e g_p} \left(\frac{m_H}{m_e m_p} \frac{2\pi}{T} \right)^{3/2} e^{\Delta E/T}, \quad (8.174)$$

which, with $m_H \approx m_p$, gives

$$\left(\frac{n_H}{n_p^2} \right)_{\text{eq}} = \left(\frac{2\pi}{m_e T} \right)^{3/2} e^{\Delta E/T}. \quad (8.175)$$

We define the free fraction,

$$X = \frac{n_p^{\text{free}}}{n_b}, \quad (8.176)$$

where $n_p = n_e$ and

$$n_b \equiv n_p + n_H \quad (8.177)$$

is the number density of baryons (ignoring the helium), with

$$n_b = \eta n_\gamma, \quad (8.178)$$

where $\eta \sim 10^{-9}$ is the baryon-to-photon ratio (Sec. 8.6). Then $n_p = X n_b$ and $n_H = (1 - X) n_b$, and, with n_γ from Eq. (8.93), this gives the *Saha equation*,

$$\frac{1 - X}{X^2} = \frac{2\zeta(3)}{\pi^2} \eta \left(\frac{2\pi T}{m_e} \right)^{3/2} e^{\Delta E/T}, \quad (8.179)$$

with $\zeta(3) \approx 1.202$. The exponential is large for $T < \Delta E$, but it is multiplied by the very small η . The hydrogen therefore remains ionised ($X \approx 1$) until T is well below ΔE , with $X = \frac{1}{2}$ at $T \approx 0.3 \text{ eV}$.

The photons are coupled to the plasma by Thomson scattering from the free electrons, with rate

$$\Gamma = n_e \sigma_T = X \eta n_\gamma \sigma_T, \quad (8.180)$$

where $\sigma_T = 8\pi\alpha^2/3m_e^2 \approx 6.65 \times 10^{-25} \text{ cm}^2$ is the Thomson cross-section. While the hydrogen is ionised, $\Gamma \gg H$. As X falls, so does Γ , and once $X \lesssim 10^{-2}$ we have $\Gamma < H$: the photons *decouple*. This is the *last scattering*, at $T \approx 0.26 \text{ eV}$ ($z \approx 1100$), shortly after recombination. Since then, the photons have travelled freely: these are the CMB, with $T = 2.7 \text{ K}$ today.

Simple estimate. Naively, we would expect recombination at $T \sim 13 \text{ eV}$, but there are so many photons,

$$n_\gamma \gg n_p, \quad (8.181)$$

that the high-energy tail of their distribution matters. The number (fraction) of photons with $E > \Delta$, where $\Delta \equiv \Delta E$, is

$$\begin{aligned} \frac{n_\gamma(E > \Delta)}{n_\gamma} &= \frac{1}{2\zeta(3)T^3} \int_\Delta^\infty \frac{E^2 dE}{e^{E/T} - 1} \\ &\approx 0.4 \left(\frac{\Delta}{T} \right)^2 e^{-\Delta/T}, \end{aligned} \quad (8.182)$$

for $T \ll \Delta$. Ionisation is efficient until there is about one such photon per baryon,

$$n_b \approx n_\gamma(E > \Delta), \quad (8.183)$$

so

$$\frac{n_b}{n_\gamma} = \eta = 0.4 \left(\frac{\Delta}{T} \right)^2 e^{-\Delta/T}, \quad (8.184)$$

where the prefactor is not important after taking the log, giving

$$T \approx \frac{-\Delta}{\ln \eta} = \frac{-13}{\ln(10^{-9})} \text{ eV} \approx 0.6 \text{ eV}. \quad (8.185)$$

This is within a factor of two of the result from the Saha equation, and far below 13 eV .

8.6 Big Bang nucleosynthesis

Big Bang nucleosynthesis (BBN) is the formation of the light nuclei from the protons and neutrons. It takes place at $T \sim 1 \text{ MeV} - 0.01 \text{ MeV}$, i.e., $1 \text{ s} - 1000 \text{ s}$. We need only the results of Sec. 8.4. The nucleons and nuclei are non-relativistic, with number densities given by Eq. (8.90), while the photons, electrons, and neutrinos are relativistic, with $n \sim T^3$ [Eq. (8.93)]. The universe is radiation

dominated, so the expansion rate and the age are given by Eqs. (8.108) and (8.110). At the start of BBN, $g_* \approx 10.75$, from $e^\pm$, γ , and $3(\bar{\nu}, \nu)$.

The main stages of BBN are summarised in Table 8.6, and the evolution of the abundances is shown in Fig. 8.11. We consider each stage in turn.

t	T	
	10 MeV	all in equilibrium (p, n)
~ 1 s	0.7 MeV	p - n freeze-out (n decay still); D produced, but quickly killed by γ
~ 180 s	0.07 MeV	D survives (“D bottleneck”); n decayed since freeze-out $\sim 10\%$; BBN starts
$\sim 10^3$ s	0.03 MeV	nuclear rates drop below H : freeze out; primordial abundances fixed; ${}^4\text{He}$ (Y_p) $\approx 25\%$; D/H strongly depends on $\eta = n_b/n_\gamma$

Table 8.6: The main stages of Big Bang nucleosynthesis.

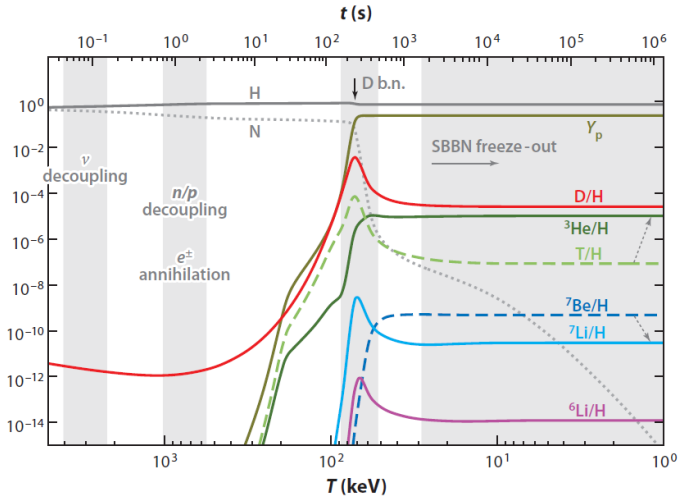


Figure 8.11: Evolution of the light-element abundances during BBN, from Ref. [7]. The neutron abundance (N) falls through the freeze-out and decay, until the deuterium bottleneck (“D b.n.”) opens at $T \approx 0.07$ MeV; the neutrons are then processed into ${}^4\text{He}$ (Y_p).

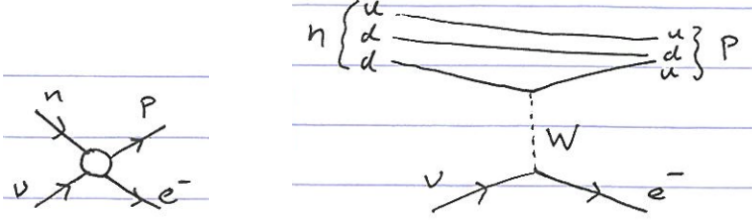


Figure 8.12: $n\nu \rightarrow pe^-$ via an effective four-point vertex, and the quark-level W -exchange diagram, $n\{u, d, d\} \rightarrow p\{u, u, d\}$ with $\nu \rightarrow e^-$.

8.6.1 p/n freeze-out

At $T \sim 10 \text{ MeV}$, the protons and neutrons are converted into each other by the weak interaction. The relevant reactions are

$$n + \nu \longleftrightarrow p + e^-, \quad (8.186)$$

$$n + e^+ \longleftrightarrow p + \bar{\nu}, \quad (8.187)$$

$$n \longleftrightarrow p + e^- + \bar{\nu} \quad (\text{decay}), \quad (8.188)$$

noting that the proton does not decay.

These keep p and n in equilibrium, so long as $\Gamma \gtrsim H$. Since $m_n \approx m_p \approx \text{GeV}$, they are non-relativistic, while the electrons, with $m_e \approx 0.5 \text{ MeV}$, are ultra-relativistic. Then, from Eq. (8.90), with $\mu_n \approx \mu_p$ (the chemical potentials of the leptons are negligible),

$$\frac{n_n}{n_p} = e^{-(m_n - m_p)/T} = e^{-\Delta/T}, \quad (8.189)$$

where $\Delta \approx 1.3 \text{ MeV}$. For $T \gg 1 \text{ MeV}$ there are equal numbers of protons and neutrons; around $T \lesssim \text{MeV}$, the lighter p is favoured.

When does this leave equilibrium? When $\Gamma \sim H$, where

$$\Gamma = n \langle \sigma v \rangle \sim G_F^2 T^5 \quad (8.190)$$

(with $n \sim T^3$ for relativistic e^-), and

$$H \sim \sqrt{\frac{8\pi^3 g_*}{90}} \frac{T^2}{m_P}, \quad (8.191)$$

which depends on the number of (light) neutrino species through g_* . These are the same weak interactions that keep the neutrinos in equilibrium, with the same

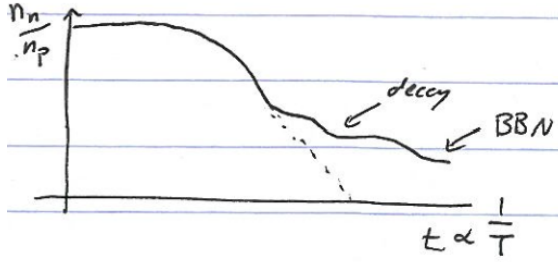


Figure 8.13: n_n/n_p against t : the equilibrium fall-off, the plateau at freeze-out, the further decrease from n decay, and the onset of BBN.

cross-section, $\sigma v \sim G_F^2 T^2$ (Sec. 8.5.1), so the p/n freeze-out happens at about the same time as the neutrino decoupling. This gives

$$T_{\text{fo}} \sim 0.7 \text{ MeV}, \quad (8.192)$$

at $t \sim 1 \text{ s}$, so that

$$\frac{n_n}{n_p} \approx e^{-1.3/0.7} = 0.16, \quad (8.193)$$

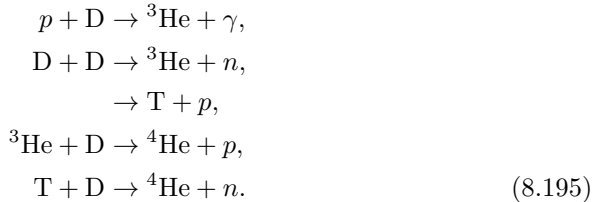
or about 1 : 6. But n keeps decaying, as shown in Fig. 8.13.

8.6.2 The deuterium bottleneck

Once the neutrons have frozen out, at $T \sim 0.7 \text{ MeV}$, the nuclei can begin to form. We write $\text{D} = {}^2\text{H}$ and $\text{T} = {}^3\text{H}$. The first step is the formation of deuterium,

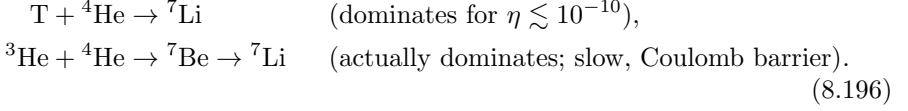


which is then processed further:



These are very fast strong interactions, and ${}^4\text{He}$ is tightly bound ($Q_{{}^4\text{He}}$ large), so all neutrons are funnelled through to ${}^4\text{He}$. Every step requires deuterium, so the outcome depends on the neutron number and on D.

There are no stable $A = 5, 8$ nuclei, so a bridge is needed to form anything heavier:



Therefore, very little is produced beyond ${}^4\text{He}$.

The abundance of deuterium follows from the equilibrium of

$$p + n \longleftrightarrow \text{D} + \gamma, \quad (8.197)$$

just as for recombination (Sec. 8.5.5). We have

$$n_i = g_i \left(\frac{m_i T}{2\pi} \right)^{3/2} e^{(\mu_i - m_i)/T} \quad (8.198)$$

for p , n , D , with the equilibrium condition

$$\mu_{\text{D}} = \mu_n + \mu_p \quad (8.199)$$

(since $\mu_\gamma = 0$). With $g_{\text{D}} = 3$, $g_p = g_n = 2$, $m_{\text{D}} \approx 2m_p \approx 2 \text{ GeV}$, and

$$m_{\text{D}} = m_p + m_n - B, \quad (8.200)$$

where $B = 2.22 \text{ MeV}$ is the binding energy of deuterium, we find

$$\frac{n_{\text{D}}}{n_n} = n_p 6 \left(\frac{m_p T}{\pi} \right)^{-3/2} e^{B/T}. \quad (8.201)$$

Here, n_p is set by the baryon-to-photon ratio,

$$\eta = \frac{n_b}{n_\gamma} \approx 3 \times 10^{-8} \Omega_b h^2 \sim 10^{-9}. \quad (8.202)$$

Initially ($\sim 10 \text{ MeV}$), $n_p \approx n_n \approx 0.5 n_b$. Since η is so small, there is very little deuterium until $T \ll B$. At low T , D is favoured over neutrons... but it can be destroyed again by the photons,

$$\text{D} + \gamma \longleftrightarrow p + n. \quad (8.203)$$

This requires high-energy photons, those with $E \gtrsim B$, of which there are [see Eq. (8.182)]

$$n_\gamma^* = n_\gamma(E \gtrsim B) \sim n_\gamma e^{-B/T}. \quad (8.204)$$

Even though $T \ll B$, there is a significant number in the high-energy tail, because there are so many more photons than baryons. This is the same effect that delays recombination (Sec. 8.5.5). D cannot survive until

$$\frac{n_\gamma^*}{n_D} \lesssim 1. \quad (8.205)$$

With $n_b = \eta n_\gamma$ and $n_\gamma^* < n_b$, this requires

$$\begin{aligned} e^{-B/T} &< \eta \\ \implies \frac{B}{T} &> \ln(1/\eta) \approx \ln(10^9), \end{aligned} \quad (8.206)$$

which gives $T \lesssim 0.1 \text{ MeV}$. This is only a rough estimate. Solving Eq. (8.201) for $n_D \sim n_n$, with $n_p \approx \eta n_\gamma$, gives

$$T \approx 0.07 \text{ MeV}, \quad (8.207)$$

which marks the start of BBN. Until then, even though the nuclear reactions are fast, no nuclei can be built, since every step requires D: this is the *deuterium bottleneck*.

The p/n freeze-out is at $T \sim 0.7 \text{ MeV}$, but BBN cannot start until $T \sim 0.07 \text{ MeV}$. Between $0.7 \rightarrow 0.07 \text{ MeV}$ ($\sim 1 \text{ s} - 180 \text{ s}$), the free neutrons decay, with lifetime

$$\tau_n \approx 880 \text{ s}, \quad (8.208)$$

so that, approximately,¹⁶

$$\left(\frac{n}{p}\right)_{\text{BBN}} \approx \left(\frac{n}{p}\right)_{\text{fo}} e^{-t/\tau_n} = \frac{1}{6} e^{-180/880} \approx \frac{1}{7}. \quad (8.209)$$

8.6.3 The primordial abundances

Once the bottleneck is passed, the reactions of Sec. 8.6.2 proceed quickly. ^4He is stable and tightly bound, so it is efficiently produced (if D is available), and essentially all n end up in ^4He . Therefore, the mass fraction of helium is

$$Y_p = \frac{4n_{\text{He}}}{n_b} \approx 4 \left(\frac{n_n/2}{n_p + n_n} \right) = \frac{2n/p}{1 + n/p}, \quad (8.210)$$

where n_p and n_n are the values before BBN, and each ^4He takes two neutrons. With $n/p \approx \frac{1}{7}$, this gives

$$Y_p \approx 25\%, \quad (8.211)$$

¹⁶We neglect the small increase in n_p from the decays.

with only a trace of deuterium left over, $D \sim 0.01\%$.

By $T \sim 0.03 \text{ MeV}$ ($\sim 10^3 \text{ s}$), the nuclear reaction rates have dropped below H , and the abundances freeze out. The details are complicated, and depend strongly on η . Large η gives faster reactions, so more ${}^4\text{He}$ and less D ; low η gives slower reactions, and so more left-over D . We have

$$\frac{D}{H} \propto \eta^{-1.6} \approx 2.6 \times 10^{-5} \left(\frac{6}{\eta_{10}} \right)^{1.6}, \quad (8.212)$$

where $\eta_{10} \equiv 10^{10}\eta$. The observed value is 2.5×10^{-5} , so $\eta_{10} = 6$, in agreement with the value found independently from the CMB. In contrast, Y_p depends only weakly on η : nearly all of the neutrons end up in ${}^4\text{He}$ whatever its value, and η enters only through the time at which the bottleneck opens, which is logarithmic. It is this strong dependence that makes D the best measure of the baryon density. Both Y_p and D/H match observations very well.

8.6.4 The lithium problem

The one exception is lithium. Recall that ${}^7\text{Li}$ is produced through the reactions of Eq. (8.196): directly, which dominates for $\eta \lesssim 10^{-10}$, and through ${}^7\text{Be}$, which later decays to ${}^7\text{Li}$, and which actually dominates for the observed η . There is a complicated dependence on η (and on τ_n). The predicted and observed abundances are

$$\left(\frac{{}^7\text{Li}}{H} \right)_{\text{pred}} \sim 5 \times 10^{-10}, \quad (8.213)$$

$$\left(\frac{{}^7\text{Li}}{H} \right)_{\text{obs}} \sim 1.5 \times 10^{-10}, \quad (8.214)$$

a $3\times$ discrepancy, known as the *lithium problem*. The pathways are complicated, and it may be that some of the lithium has since been destroyed; the question remains open. The predicted abundances are compared with the observations in Fig. 8.14.

8.7 Dark matter

About 85% of the matter in the universe is *dark matter*: it does not emit or absorb light, and is known only through its gravity. The evidence for it is very strong, and comes from many independent observations, on very different scales. For the history, see Bertone and Hooper¹⁷. We saw in Sec. 8.4.5 how it may have been produced; here, we briefly review the evidence, the candidates, and how we might detect it.

¹⁷G. Bertone and D. Hooper, *A History of Dark Matter*, Rev. Mod. Phys. **90**, 045002 (2018).

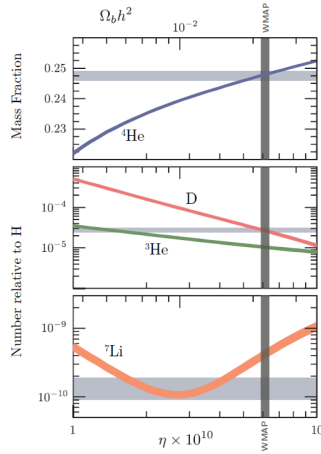


Figure 8.14: The ratios of light elements very sensitive to baryon density. The coloured curves show the theory predictions for Y_p (${}^4\text{He}$), D/H , and ${}^7\text{Li}/\text{H}$ against η , with the observed abundances in light grey, and the observed η band in dark grey.

8.7.1 Evidence and properties

The first major hints came from galaxy clusters. Zwicky found that the galaxies of the Coma cluster move much too fast to be held together by the visible matter, and coined the term “dark matter”.

The same is seen within galaxies, in the *galactic rotation curves* measured by Rubin and Ford. In all galaxies, the stars and gas far from the centre orbit faster than can be explained by the visible matter. This is not a small correction: 80–95% of the mass is “missing”. It implies either extra, unseen mass,

$$a = \frac{G [M(r) + \delta M(r)]}{r^2}, \quad (8.215)$$

which simulations and observations show forms a *halo* around the galaxy, or else a modification of gravity,

$$a = \frac{[G + \delta G(M, r)] M(r)}{r^2}. \quad (8.216)$$

The *Bullet Cluster*, a pair of colliding galaxy clusters, distinguishes between the two. Most of the ordinary matter is hot gas, which is seen in X-rays, and which is slowed by the collision. The mass, however, which is mapped by gravitational lensing, has passed straight through: it is not where the gas is. This is what we expect for frictionless dark matter, and is hard to explain by a modification of gravity alone.

As we saw in Sec. 8.6.3, Big Bang nucleosynthesis is very sensitive to the density of baryons, Ω_b . Combined with other measurements of the total matter density (e.g., from galaxy surveys), this shows that $\Omega_m \gg \Omega_b$: the dark matter cannot be made of baryons.

The temperature anisotropies of the CMB are very sensitive to the amounts of both ordinary and dark matter. Roughly, the baryons interact with the photons, while all of the matter clumps under gravity, so that the relative heights of the peaks in the power spectrum give the ratio of baryons to dark matter. This gives the densities of Table 8.2, with $\Omega_m = \Omega_b + \Omega_c$.

There is further evidence from gravitational lensing, from the large-scale structure of the universe (its “clumpiness”), and from baryon acoustic oscillations. The Λ CDM model (Sec. 8.2.4), with a single cold dark matter component, describes all of these extremely well.

It is not unusual for a particle not to interact with light; neutrinos, for example, do not. The simplest explanation is therefore that dark matter is a new particle. From the evidence above, it must be massive, dark (no electromagnetic interactions), non-baryonic, “cold” (non-relativistic, to allow structure to form), stable on cosmological timescales, and produced with the right relic abundance in the early universe. No particle of the Standard Model satisfies all of these.

Beyond this, very little is known. In particular, the range of possible masses is enormous. For a fundamental particle,

$$m_\chi \gtrsim 10^{-21} \text{ eV}, \quad (8.217)$$

$$m_\chi \lesssim 10^{28} \text{ eV}, \quad (8.218)$$

where the lower limit comes from requiring that the de Broglie wavelength fits inside a galaxy, and the upper limit is the Planck mass: a range of 50 orders of magnitude. Including composite objects, up to black holes, it is 90 orders of magnitude.

8.7.2 Candidates

Some options are disfavoured: primordial black holes are strongly constrained by experiment, massive compact halo objects (MACHOs) are baryonic, and the neutrinos of the Standard Model are too light (“warm”), with too small a relic density. Good candidates include weakly interacting massive particles (WIMPs), axions and axion-like particles, and sterile neutrinos. The vast majority of the focus has been on WIMPs, but the field is very wide.

A WIMP (Sec. 8.4.5) is a new particle with a mass of $\sim 10 \text{ GeV}$ to $\sim 10 \text{ TeV}$, which interacts only through the weak interaction (and gravity). It is dark, since it has no electromagnetic interactions; cold, since it is very massive; and non-baryonic and stable, so that it causes no problems for BBN or the CMB. Most

importantly, as we saw, a weak-scale annihilation cross-section naturally leads to a relic abundance of the right order of magnitude: the WIMP miracle.

This is an example of *thermal* production: the dark matter begins in thermal equilibrium with the other particles in the early universe, and its abundance is fixed when it falls out of equilibrium. In *non-thermal* production (Sec. 8.7.3), the relic density is set by some other process.

The *axion* was proposed to solve a different problem. The Standard Model allows a *CP*-violating term in the strong interaction, with a parameter, θ , which is an angle, and could take any value. It would produce an electric dipole moment of the neutron, which is not observed, so that $\theta < 10^{-10}$: this is the *strong CP problem*. The axion is a new field that dynamically drives θ to zero. Axions are very light ($\ll \text{eV}$), so their number density must be very high, and they behave as a classical field; they may be the cold dark matter. The axion is therefore an attractive candidate, since it would solve two problems at once.

8.7.3 Non-thermal production

Not all dark matter candidates are thermal relics. If the dark matter interacts so feebly that it never reaches equilibrium, it can instead be produced slowly from the thermal bath, until the production stops: this is called *freeze-in*. Other possibilities include the decay of a heavier particle. The axion is produced in a different way again, by *misalignment*. The axion field starts away from the minimum of its potential, and is held there by the expansion [the friction term in Eq. (8.232)] until $H \sim m_a$. It then oscillates about the minimum, and these oscillations behave as cold matter, $\rho \propto a^{-3}$, even though the particle is very light.

8.7.4 Direct detection

If dark matter interacts with the Standard Model particles other than through gravity, there are three ways to look for it. These correspond to the same diagram read in three directions (Fig. 8.15): scattering (*direct detection*), annihilation (*indirect detection*), and production at colliders.

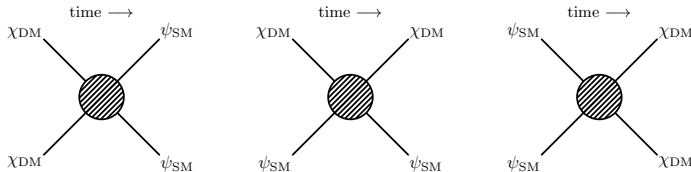


Figure 8.15: The three ways to search for dark matter: annihilation (indirect detection), scattering (direct detection), and production (colliders).

The Solar System lies within the dark matter halo of the Galaxy, so there is

a constant flux of dark matter particles through any detector on Earth. In the *standard halo model*, the local density is $\rho_{\text{DM}} \approx 0.4 \text{ GeV/cm}^3$, and the velocities follow a Maxwell-Boltzmann distribution with typical speed $v_0 \approx 235 \text{ km/s}$, cut off at the escape velocity, $v_{\text{esc}} \approx 550 \text{ km/s}$. Note that $v \sim 10^{-3}$ (in units of c): the dark matter is very non-relativistic.

In direct detection, we look for the recoil of a nucleus struck by one of these particles. From Eq. (8.70), the event rate is

$$R = N_{\text{target}} \frac{\rho_{\text{DM}}}{m_\chi} \langle \sigma v \rangle, \quad (8.219)$$

where N_{target} is the number of target nuclei, ρ_{DM}/m_χ is the number density of the dark matter, and the average is over the velocity distribution. The recoil energy is very small. For elastic scattering from a nucleus of mass m_N , the recoil energy is at most

$$E_{\text{max}} = \frac{2\mu^2 v^2}{m_N}, \quad (8.220)$$

where $\mu = m_\chi m_N / (m_\chi + m_N)$ is the reduced mass. For $m_\chi \sim m_N \sim 100 \text{ GeV}$, this is of order tens of keV. The detectors must therefore have a very low energy threshold, and very low backgrounds: they are built from radio-pure materials, and placed deep underground to shield them from cosmic rays.

What is measured is the rate as a function of the recoil energy. This depends on the dark matter mass, the cross-section, and the local density and velocity distribution, which cannot be separated in a single measurement. A light particle cannot give a recoil above the threshold, while for a heavy particle the number density, ρ_{DM}/m_χ , is small; the sensitivity is therefore best for $m_\chi \sim m_N$. The cross-section depends on how the dark matter couples to the nucleus. If the scattering is *spin-independent*, it is coherent over all of the nucleons, so that $\sigma \propto A^2$, which strongly favours heavy nuclei. If it is *spin-dependent*, only the unpaired nucleons contribute, and the nucleus must have a net spin. To compare different experiments, the results are quoted as a cross-section per nucleon.

How would we know that a signal is not just noise? The Sun moves through the halo, so we see a “WIMP wind”, and the orbital velocity of the Earth adds to, or subtracts from, this through the year. The rate should therefore show an *annual modulation*, of about 5%, and a preferred direction. It also helps to detect the recoil in more than one way. A recoiling nucleus produces ionisation, scintillation light, and vibrations (phonons), in different proportions to the background events. The most sensitive experiments (XENON, LZ, PandaX) use large tanks of liquid xenon, and measure both the light and the ionisation; this also gives the position of each event, so that those near the walls can be rejected.

8.7.5 Indirect detection

In indirect detection, we look for the Standard Model particles produced when dark matter annihilates (or decays) in space: gamma rays, neutrinos, and anti-matter. The same annihilation process that set the relic abundance in the early universe continues today, in regions where the dark matter density is high. This probes the annihilation cross-section directly, to be compared with that required for a thermal relic, $\langle\sigma v\rangle \approx 3 \times 10^{-26} \text{ cm}^3/\text{s}$ (Sec. 8.4.5).

Gamma rays and neutrinos are the simplest messengers, since they travel in straight lines, and point back to their source. The annihilation rate per unit volume is $\langle\sigma v\rangle n_\chi^2$ [Eq. (8.72)], so the flux from a given direction is

$$\Phi \propto \frac{\langle\sigma v\rangle}{m_\chi^2} J, \quad (8.221)$$

where

$$J \equiv \int_{\Delta\Omega} \int_{\text{l.o.s.}} \rho_{\text{DM}}^2 \, dl \, d\Omega \quad (8.222)$$

is the integral along the line of sight, over the region observed. The first factor depends on the particle physics, and the second, known as the J -factor, on the astrophysics. The best targets are therefore regions with a high dark matter density, and few other sources of gamma rays. The Galactic centre gives a large signal, but has a large background; the dwarf galaxies give a small signal, but have very little background.

The gamma rays may be produced directly, $\chi\bar{\chi} \rightarrow \gamma\gamma$, which would give a sharp line at $E_\gamma = m_\chi$: a very clear signal, though the rate is expected to be small. More often, they come from the decays of the other annihilation products, which give a broad spectrum. Observations of dwarf galaxies exclude the thermal cross-section for $m_\chi \lesssim 100 \text{ GeV}$, though this depends on the model. An excess of GeV gamma rays is observed from the Galactic centre; it may be a dark matter signal, but is more likely due to pulsars.

Neutrinos provide a complementary probe. Dark matter particles passing through the Sun occasionally scatter, lose energy, and become gravitationally bound. They sink to the core, where they annihilate, and the neutrinos produced can be seen on Earth by detectors such as IceCube. Since the Sun is mostly hydrogen, this is sensitive to the spin-dependent scattering cross-section, and is competitive with direct detection.

Charged particles (positrons and antiprotons) are harder to interpret: they are deflected by the Galactic magnetic fields, so they do not point back to their source. An excess of positrons over the expected background is observed (PAMELA, AMS-02), but it too may be due to pulsars.

Finally, dark matter may be produced in high-energy collisions, where it would escape the detector, and show up as “missing” energy and momentum. This does

not depend on the dark matter in the Galaxy at all; but, equally, a new particle found in this way need not be the dark matter.

8.8 Open problems

The picture developed above, the Λ CDM model with a hot Big Bang, passes many demanding tests, but it is incomplete. It does not explain the initial conditions: why there is more matter than antimatter, and why the universe began so uniform and flat. We do not know what the dark matter is (Sec. 8.7), nor the dark energy. There are also tensions within the model itself, such as the lithium problem (Sec. 8.6.4), and the expansion rate today, H_0 , for which the value measured locally is larger than that inferred from the CMB (the “Hubble tension”). For a review of other, less widely discussed, anomalies, see Peebles¹⁸.

8.8.1 Baryogenesis

We observe

$$\eta = \frac{n_B - n_{\bar{B}}}{n_\gamma} \sim 6 \times 10^{-10}, \quad (8.223)$$

from both BBN (Sec. 8.6.3) and the CMB. Why is this not zero? If there had been equal numbers of baryons and antibaryons, they would have annihilated almost completely, leaving essentially no matter at all.

Is it just the initial conditions? Maybe, but why? Moreover, inflation should wash this out (Sec. 8.8.2): any initial asymmetry is massively diluted, and reheating then produces a large density of new particles. The asymmetry must therefore have been generated afterwards; this is called *baryogenesis*.

What could produce this? There are three requirements, the *Sakharov conditions*:

1. *B violation*. There must be processes that change the baryon number. For $X \rightarrow Y, Z$, with B the baryon number of the initial state and B' that of the final state, we require $B' \neq B$.
2. *C and CP violation* (C swaps particles with antiparticles, and P is a mirror reflection). Otherwise, the conjugate process, $\bar{X} \rightarrow \bar{Y} \bar{Z}$, would produce the opposite baryon number at the same rate, with no net effect. We require

$$R(X \rightarrow YZ) \neq R(\bar{X} \rightarrow \bar{Y} \bar{Z}). \quad (8.224)$$

3. *Departure from thermal equilibrium*. We require

$$R(X \rightarrow YZ) \neq R(YZ \rightarrow X) \quad (8.225)$$

¹⁸P. J. E. Peebles, *Anomalies in Physical Cosmology*, *Annals Phys.* **447**, 169159 (2022).

(and inverse). In equilibrium, the inverse processes undo any asymmetry, and

$$\frac{n_X}{n_{\bar{X}}} \sim \exp(-(E_X - E_{\bar{X}})/T) = 1, \quad (8.226)$$

by detailed balance, since a particle and its antiparticle have the same mass. So we require

$$R(YZ \rightarrow X) \lesssim H, \quad (8.227)$$

where H is the expansion rate of the universe – noting that the decay still occurs.

All three are possible in the Standard Model. In particular, B violation is allowed (it is “anomalous”: a quantum effect, which is important only above the electroweak scale), though it has never been observed. However, the CP violation of the Standard Model is far too small to give the observed η , and the departure from equilibrium is too weak. Baryogenesis therefore requires new sources of CP violation, and so physics beyond the Standard Model.

Two well-motivated possibilities are *electroweak baryogenesis* and *leptogenesis*. In the first, the asymmetry is generated at the electroweak phase transition, using the B violation of the Standard Model itself. This requires new particles near the electroweak scale, both to make the transition sufficiently far from equilibrium and to provide the CP violation. It can therefore be tested in the laboratory, for example in searches for electric dipole moments. In the second, the out-of-equilibrium, CP -violating decays of new heavy neutrinos first produce an excess of leptons over antileptons. The B -violating processes of the Standard Model, which also violate lepton number, then convert part of this into an excess of baryons.

8.8.2 Inflation

The CMB has the same temperature across the whole sky, even between patches that, in a universe always dominated by radiation or matter, could never have been in causal contact. This is the *horizon problem*. The distance a signal can travel in an expansion time is the Hubble radius, $1/H$. In comoving coordinates this is $1/aH = 1/\dot{a}$, which grows if the expansion decelerates, so that we are always seeing new regions for the first time. The problem is solved if there was an early period of *accelerated* expansion, called *inflation*: the comoving Hubble radius then shrinks, and everything we now observe comes from a single region that was once in causal contact. Inflation also drives the curvature term in Eq. (8.48) to zero, and dilutes whatever was present before it.

For a flat universe, the acceleration equation, Eq. (8.49), can be written

$$\frac{\ddot{a}}{a} = -\frac{H^2}{2} (1 + 3w), \quad (8.228)$$

where $w = P/\rho$. Inflation therefore requires $w < -\frac{1}{3}$, which neither matter nor radiation can provide.

The simplest way to obtain $w < -\frac{1}{3}$ is with a scalar field, ϕ , called the *inflaton*, with potential $V(\phi)$. Since the universe is homogeneous, we immediately discard the spatial derivatives, $\phi = \phi(t)$, so that the Lagrangian density is

$$\mathcal{L} = \frac{1}{2}\dot{\phi}^2 - V(\phi). \quad (8.229)$$

The energy density and pressure follow from the stress-energy tensor (see Fields Sec. 2.5):

$$\rho = \frac{1}{2}\dot{\phi}^2 + V(\phi), \quad (8.230)$$

$$P = \frac{1}{2}\dot{\phi}^2 - V(\phi). \quad (8.231)$$

If the potential energy dominates, $\dot{\phi}^2 \ll V$, then $w \approx -1$: the field behaves like a cosmological constant, and the expansion is (almost) exponential, $a \propto e^{Ht}$.

The equation of motion follows from the fluid equation, Eq. (8.50), with $\rho + P = \dot{\phi}^2$, and the expansion rate from the Friedmann equation, Eq. (8.105):

$$\ddot{\phi} + 3H\dot{\phi} + V' = 0, \quad (8.232)$$

$$H^2 = \frac{8\pi}{3m_P^2} \left(\frac{1}{2}\dot{\phi}^2 + V \right), \quad (8.233)$$

where $V' \equiv dV/d\phi$. The field rolls down its potential, with the expansion acting as friction. In the *slow-roll* approximation, we take $\dot{\phi}^2 \ll V$ and $\ddot{\phi} \ll 3H\dot{\phi}$, so that

$$3H\dot{\phi} \approx -V', \quad (8.234)$$

$$H^2 \approx \frac{8\pi}{3m_P^2} V. \quad (8.235)$$

Combining these, we find $\dot{\phi}^2/V \approx \frac{2}{3}\epsilon$, where

$$\epsilon \equiv \frac{m_P^2}{16\pi} \left(\frac{V'}{V} \right)^2. \quad (8.236)$$

Therefore, the approximation holds, and the universe inflates, so long as $\epsilon \ll 1$: the potential must be flat.¹⁹ Inflation ends when $\epsilon \sim 1$, i.e., when $\dot{\phi}^2 \sim V$.

¹⁹Neglecting $\ddot{\phi}$ also requires that $m_P^2 |V''|/8\pi V \ll 1$.

The amount of inflation is measured by the number of *e-folds*, $N \equiv \ln(a_{\text{end}}/a_i)$. Using Eqs. (8.234) and (8.235),

$$\begin{aligned} N &= \int_{t_i}^{t_{\text{end}}} H \, dt = \int_{\phi_i}^{\phi_{\text{end}}} \frac{H}{\dot{\phi}} \, d\phi \\ &\approx \frac{8\pi}{m_P^2} \int_{\phi_{\text{end}}}^{\phi_i} \frac{V}{V'} \, d\phi. \end{aligned} \quad (8.237)$$

Solving the horizon problem requires $N \gtrsim 60^{20}$. As an example, take $V = \frac{1}{2}m^2\phi^2$. Then $\epsilon = m_P^2/4\pi\phi^2$, so inflation ends at $\phi_{\text{end}} \approx m_P/\sqrt{4\pi}$, and

$$N \approx \frac{2\pi}{m_P^2} (\phi_i^2 - \phi_{\text{end}}^2) = \frac{2\pi\phi_i^2}{m_P^2} - \frac{1}{2}. \quad (8.238)$$

Sufficient inflation therefore requires that the field starts at $\phi_i \gtrsim 3m_P$.

After inflation ends, the field oscillates around the minimum of its potential. Through its coupling to the Standard Model particles, it decays, filling the universe with a hot plasma of photons and particles: this is *reheating*, and is the start of the hot Big Bang. Since inflation dilutes anything that was present before it, all of the matter we observe – including the baryon asymmetry (Sec. 8.8.1) – must have been produced during or after reheating.

Perturbations. Inflation also provides an origin for the structure in the universe. Quantum fluctuations of the inflaton are stretched by the expansion to scales larger than the Hubble radius, where they become classical perturbations of the density: the seeds of the CMB anisotropies and, later, of the galaxies. Since H is almost constant during slow roll, the perturbations have almost the same amplitude on all scales: the spectrum is nearly *scale invariant*, $\propto k^{n_s-1}$, with n_s slightly less than one. For the quadratic potential, $n_s \approx 1 - 2/N \approx 0.97^{21}$, to be compared with the Planck value²²,

$$n_s = 0.965 \pm 0.004. \quad (8.239)$$

Inflation also predicts primordial gravitational waves, which have not been observed; the quadratic potential predicts too many, so it serves here only as an illustration.

²⁰D. Baumann, *TASI lectures on inflation*, arXiv:0907.5424.

²¹D. Baumann, *TASI lectures on inflation*, arXiv:0907.5424.

²²Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, Astron. Astrophys. **641**, A6 (2020).

8.8.3 Dark energy

About 70% of the energy density of the universe today is *dark energy* (Table 8.2): a component with negative pressure, $w < -\frac{1}{3}$, which causes the expansion to accelerate (Sec. 8.2.2).

The first direct evidence came from type Ia supernovae. These are “standard candles”, so that their brightness gives their distance, and distant supernovae are fainter than expected in a decelerating universe. Independently, the CMB shows that the universe is flat, $\Omega_{\text{total}} \approx 1$, while the matter makes up only $\Omega_m \approx 0.3$: the remainder must be a component that does not cluster.

All of the observations are consistent with a cosmological constant, $w = -1$, with [using Eq. (8.109)]

$$\rho_\Lambda = \Omega_\Lambda \rho_{\text{crit}} \approx (2 \text{ meV})^4. \quad (8.240)$$

The problem is to explain this value. A cosmological constant is equivalent to an energy density of the vacuum, and the natural scale for that in quantum field theory is m_P^4 , which is too large by some 120 orders of magnitude. This is the *cosmological constant problem*²³. The alternatives are that the dark energy is dynamical, for example a slowly rolling scalar field like the inflaton (in which case $w \neq -1$, and changes with time), or that gravity is modified on the largest scales.

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